Red Hat OpenShift AI Self-Managed 3.0

Working with AI pipelines

Work with AI pipelines from Red Hat OpenShift AI Self-Managed

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Legal Notice

Abstract

Enhance your projects on OpenShift AI by building portable machine learning (ML) workflows with AI pipelines.

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PREFACE

As a data scientist, you can enhance your projects on OpenShift AI by building portable machine learning (ML) workflows with AI pipelines, using Docker containers. This enables you to standardize and automate machine learning workflows to enable you to develop and deploy your data science models.

For example, the steps in a machine learning workflow might include items such as data extraction, data processing, feature extraction, model training, model validation, and model serving. Automating these activities enables your organization to develop a continuous process of retraining and updating a model based on newly received data. This can help address challenges related to building an integrated machine learning deployment and continuously operating it in production.

You can also use the Elyra JupyterLab extension to create and run AI pipelines within JupyterLab. For more information, see Working with pipelines in JupyterLab.

To use an AI pipeline in OpenShift AI, you need the following components:

- **Pipeline server**: A server that is attached to your project and hosts your AI pipeline.
- **Pipeline**: A pipeline defines the configuration of your machine learning workflow and the relationship between each component in the workflow.
  - Pipeline code: A definition of your pipeline in a YAML file.
  - Pipeline graph: A graphical illustration of the steps executed in a pipeline run and the relationship between them.
- **Pipeline experiment**: A workspace where you can try different configurations of your pipelines. You can use experiments to organize your runs into logical groups.
  - Archived pipeline experiment: An archived pipeline experiment.
  - Pipeline artifact: An output artifact produced by a pipeline component.
  - Pipeline execution: The execution of a task in a pipeline.
- **Pipeline run**: An execution of your pipeline.
  - Active run: A pipeline run that is executing, or stopped.
  - Scheduled run: A pipeline run that is scheduled to execute at least once.
  - Archived run: An archived pipeline run.

This feature is based on Kubeflow Pipelines 2.0. Use the latest Kubeflow Pipelines 2.0 SDK to build your pipeline in Python code. After you have built your pipeline, use the SDK to compile it into an Intermediate Representation (IR) YAML file. The OpenShift AI user interface enables you to track and manage pipelines, experiments, and pipeline runs. To view a record of previously executed, scheduled, and archived runs, you can go to **Develop & train** > **Pipelines** → **Runs**, or you can select an experiment from the **Develop & train** → **Experiments** page to access all of its pipeline runs. You can manage incremental changes to pipelines in OpenShift AI by using versioning. This allows you to develop and deploy pipelines iteratively, preserving a record of your changes.

You can store your pipeline artifacts in an S3-compatible object storage bucket so that you do not consume local storage. To do this, you must first configure write access to your S3 bucket on your storage account.

CHAPTER 1. MANAGING AI PIPELINES

1.1. CONFIGURING A PIPELINE SERVER

Before you can successfully create an AI pipeline in OpenShift AI, you must configure a pipeline server. This task includes configuring where your pipeline artifacts and data are stored.

NOTE

You are not required to specify any storage directories when configuring a connection for your pipeline server. When you import a pipeline, the /**pipelines** folder is created in the **root** folder of the bucket, containing a YAML file for the pipeline. If you upload a new version of the same pipeline, a new YAML file with a different ID is added to the /**pipelines** folder.

When you run a pipeline, the artifacts are stored in the /**pipeline-name** folder in the **root** folder of the bucket.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project that you can add a pipeline server to.
- You have an existing S3-compatible object storage bucket and you have configured write access to your S3 bucket on your storage account.
- If you are configuring a pipeline server for production pipeline workloads, you have an existing external MySQL or MariaDB database.
- If you are configuring a pipeline server with an external MySQL database, your database must use at least MySQL version 5.x. However, Red Hat recommends that you use MySQL version 8.x.

NOTE

The **mysql_native_password** authentication plugin is required for the ML Metadata component to successfully connect to your database. **mysql_native_password** is disabled by default in MySQL 8.4 and later. If your database uses MySQL 8.4 or later, you must update your MySQL deployment to enable the **mysql_native_password** plugin.

For more information about enabling the **mysql_native_password** plugin, see Native Pluggable Authentication in the MySQL documentation.

- If you are configuring a pipeline server with a MariaDB database, your database must use MariaDB version 10.3 or later. However, Red Hat recommends that you use at least MariaDB version 10.5.

Procedure

1. From the OpenShift AI dashboard, click **Projects**.

2. On the **Projects** page, click the name of the project that you want to configure a pipeline server for.
   The project details page opens.

3. Click the **Pipelines** tab.

4. Click **Configure pipeline server**.
   The **Configure pipeline server** dialog opens.

5. In the **Object storage connection** section, provide values for the mandatory fields:

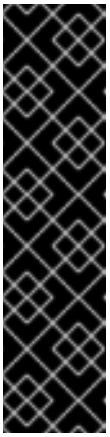
   a. In the **Access key** field, enter the access key ID for the S3-compatible object storage provider.

   b. In the **Secret key** field, enter the secret access key for the S3-compatible object storage account that you specified.

   c. In the **Endpoint** field, enter the endpoint of your S3-compatible object storage bucket.

   d. In the **Region** field, enter the default region of your S3-compatible object storage account.

   e. In the **Bucket** field, enter the name of your S3-compatible object storage bucket.



IMPORTANT

If you specify incorrect connection settings, you cannot update these settings on the same pipeline server. Therefore, you must delete the pipeline server and configure another one.

If you want to use an existing artifact that was not generated by a task in a pipeline, you can use the kfp.dsl.importer component to import the artifact from its URI. You can only import these artifacts to the S3-compatible object storage bucket that you define in the **Bucket** field in your pipeline server configuration. For more information about the **kfp.dsl.importer** component, see Special Case: Importer Components .

6. Click **Advanced settings** to display the **Database**, **Pipeline definition storage**, and **Pipeline caching** sections.

7. In the **Database** section, choose one of the following options to specify where to store your pipeline metadata and run information:

   - Select **Default database on the cluster** to deploy a MariaDB database in your project.

IMPORTANT

The **Default database on the cluster** option is intended for development and testing purposes only. For production pipeline workloads, select the **External MySQL database** option to use an external MySQL or MariaDB database.

   - Select **External MySQL database** to add a new connection to an external MySQL or MariaDB database that your pipeline server can access.

     i. In the **Host** field, enter the database hostname.

ii. In the **Port** field, enter the database port.

iii. In the **Username** field, enter the default user name that is connected to the database.

iv. In the **Password** field, enter the password for the default user account.

v. In the **Database** field, enter the database name.

8. Optional: By default, pipeline definitions are stored as Kubernetes resources, enabling version control, GitOps workflows, and integration with OpenShift GitOps or similar tools. To store pipeline definitions in the internal database instead, clear the **Store pipeline definitions in Kubernetes** checkbox in the **Pipeline definition storage** section.

9. Optional: By default, caching is configurable at both the pipeline and task levels. To disable caching for all pipelines and tasks in the pipeline server and override any pipeline-level and task-level caching settings, clear the **Allow caching to be configured per pipeline and task** checkbox in the **Pipeline caching** section.

10. Click **Configure pipeline server**.

Verification

On the **Pipelines** tab for the project:

- The **Import pipeline** button is available.
- When you click the action menu (⋮) and then click **Manage pipeline server configuration**, the pipeline server details are displayed.

Additional resources

- Overview of pipelines caching

1.1.1. Configuring a pipeline server with an external Amazon RDS database

To configure a pipeline server with an external Amazon Relational Database Service (RDS) database, you must configure OpenShift AI to trust the certificates issued by its certificate authorities (CA).

IMPORTANT

If you are configuring a pipeline server for production pipeline workloads, Red Hat recommends that you use an external MySQL or MariaDB database.

Prerequisites

- You have cluster administrator privileges for your OpenShift cluster.
- You have logged in to Red Hat OpenShift AI.
- You have created a project that you can add a pipeline server to.
- You have an existing S3-compatible object storage bucket, and you have configured your storage account with write access to your S3 bucket.

Procedure

1. Before configuring your pipeline server, from Amazon RDS: Certificate bundles by AWS Region , download the PEM certificate bundle for the region that the database was created in. For example, if the database was created in the **us-east-1** region, download **us-east-1-bundle.pem**.

2. In a terminal window, log in to the OpenShift cluster where OpenShift AI is deployed.

```
oc login api.<cluster_name>.<cluster_domain>:6443 --web
```

3. Run the following command to fetch the current OpenShift AI trusted CA configuration and store it in a new file:

```
oc get dscinitializations.dscinitialization.opendatahub.io default-dsci -o json | jq
'.spec.trustedCABundle.customCABundle' > /tmp/my-custom-ca-bundles.crt
```

4. Run the following command to append the PEM certificate bundle that you downloaded to the new custom CA configuration file:

```
cat us-east-1-bundle.pem >> /tmp/my-custom-ca-bundles.crt
```

5. Run the following command to update the OpenShift AI trusted CA configuration to trust certificates issued by the CAs included in the new custom CA configuration file:

```
oc patch dscinitialization default-dsci --type='json' -
p='[{"op":"replace","path":"/spec/trustedCABundle/customCABundle","value":"""$(awk '{printf
"%s\\n", $0}' /tmp/my-custom-ca-bundles.crt)"""}]'
```

6. Configure a pipeline server, as described in Configuring a pipeline server.

Verification

- The pipeline server starts successfully.
- You can import and run AI pipelines.

1.2. DEFINING A PIPELINE

The Kubeflow Pipelines SDK enables you to define end-to-end machine learning and AI pipelines. Use the latest Kubeflow Pipelines 2.0 SDK to build your AI pipeline in Python code. After you have built your pipeline, use the SDK to compile it into an Intermediate Representation (IR) YAML file. For more information about compiling pipelines, see *Compiling the pipeline YAML with the Kubeflow Pipelines SDK* and *Compiling Kubernetes-native manifests with the Kubeflow Pipelines SDK* . Compiling to Kubernetes-native manifests is optional and applies only when your pipeline server is configured to use Kubernetes API storage. After defining the pipeline, you can import the YAML file to the OpenShift AI dashboard to enable you to configure its execution settings.

IMPORTANT

If you are using OpenShift AI on a cluster running in FIPS mode, any custom container images for AI pipelines must be based on UBI 9 or RHEL 9. This ensures compatibility with FIPS-approved pipeline components and prevents errors related to mismatched OpenSSL or GNU C Library (glibc) versions.

You can also use the Elyra JupyterLab extension to create and run AI pipelines within JupyterLab. For more information about creating pipelines in JupyterLab, see Working with pipelines in JupyterLab. For more information about the Elyra JupyterLab extension, see Elyra Documentation.

Additional resources

- Kubeflow Pipelines 2.0 Documentation
- Elyra Documentation

1.2.1. Compiling the pipeline YAML with the Kubeflow Pipelines SDK

Before you can define your pipeline in the cluster, you must convert your Python-defined pipeline into YAML format. You can use the Kubeflow Pipelines (KFP) Software Development Kit (SDK) to compile your pipeline code into a deployable YAML file for declarative GitOps deployment.

Prerequisites

- You have installed Python 3.11 or later in your local environment.
- You have installed the Kubeflow Pipelines SDK package (**kfp**) version 2.14.3 or later.
- You have a valid Python pipeline definition file.

Procedure

Compile your pipeline by using the KFP SDK to generate the pipeline YAML file.

In the following example, replace *<pipeline_file>.py* with the name of your Python pipeline file and specify an output file for the compiled YAML:

```
$ kfp dsl compile \
   --py <pipeline_file>.py \
   --output <compiled_pipeline_file>.yaml
```

NOTE

The generated **<compiled_pipeline_file>.yaml** file contains the compiled pipeline specification in YAML format. You can use this content as the value of the **pipelineSpec** field when you create a **PipelineVersion** custom resource (CR). You can also store the file in Git for declarative or GitOps-based deployment.

Verification

Verify that the generated file includes a **pipelineSpec** key followed by the compiled pipeline definition:

```
$ head -n 10 <compiled_pipeline_file>.yaml
```

Additional resources

- Compiling a pipeline with the Kubeflow Pipelines SDK

1.2.2. Compiling Kubernetes-native manifests with the Kubeflow Pipelines SDK

If your pipeline server uses the Kubernetes native API mode, you can compile your pipeline directly to Kubernetes manifests. The output includes **Pipeline** and **PipelineVersion** custom resources with **spec.pipelineSpec** and, when you use Kubernetes resource configuration, an optional **spec.platformSpec**.

Prerequisites

- You have installed Python 3.11 or later in your local environment.
- You have installed the Kubeflow Pipelines SDK package (**kfp**) version 2.14.3 or later.
- You have a valid Python pipeline definition file.

Procedure

1. Save the following code as a new file named **compile.py** in your working directory. The example uses the **KubernetesManifestOptions** class from the **kfp.compiler.compiler_utils** module to define pipeline metadata such as the name, version, and namespace.

   #### Example compile script

   ```
   from kfp import dsl, compiler
   from kfp.compiler.compiler_utils import KubernetesManifestOptions

   @dsl.pipeline(name="<pipeline_name>")
   def my_pipeline():
       pass  # define your tasks

   compiler.Compiler().compile(
       pipeline_func=my_pipeline,
       package_path="<output_file>.yaml",
       kubernetes_manifest_format=True,
       kubernetes_manifest_options=KubernetesManifestOptions(
           pipeline_name="<pipeline_name>",
           pipeline_version_name="<version_name>",
           namespace="<namespace>",
           include_pipeline_manifest=True,
       ),
   )
   ```

2. Run the script to compile your pipeline and generate the Kubernetes manifests:

   ```
   $ python compile.py
   ```

Verification

Verify that the compiled output includes the expected resources:

```
apiVersion: pipelines.kubeflow.org/v2beta1
kind: Pipeline
---
apiVersion: pipelines.kubeflow.org/v2beta1
kind: PipelineVersion
```

```
spec:
  pipelineSpec: ...
  platformSpec: ...   # present when Kubernetes resource configuration is used
```

Additional resources

- Compiling for Kubernetes native API mode

1.2.3. Authenticating the Kubeflow Pipelines SDK with a pipeline server

You can connect the Kubeflow Pipelines (KFP) SDK to a pipeline server that is exposed by OpenShift AI. The pipeline server route is protected by OpenShift OAuth, so you must provide a valid access token when you create the KFP client.

Prerequisites

- You have logged in to the OpenShift CLI (**oc**) as a user who can access the project.
- You have created a project and configured a pipeline server for that project.
- You have installed Python and the required packages in your environment.
- Optional: If your cluster uses a custom or self-signed certificate, you know the path to the trusted certificate bundle that your environment uses.

Procedure

1. Set environment variables for your project and pipeline server route:

```
export NAMESPACE=<project_namespace>
export DSPA_NAME=$(oc -n "$NAMESPACE" get dspa -o
jsonpath='{.items[0].metadata.name}')
export API_URL="https://$(oc -n "$NAMESPACE" get route "ds-pipeline-${DSPA_NAME}" -o
jsonpath='{.spec.host}')"
```

   Replace **<project_namespace>** with the name of your project.

2. Obtain an OpenShift access token for the current user:

```
export OCP_TOKEN=$(oc whoami --show-token)
```

NOTE

Avoid pasting the access token directly into commands or scripts. The token can appear in your shell history or in process listings if you pass it as a literal argument.

To reduce this risk, store the token in an environment variable and reference it from your code or commands. For example:

```
./.venv/bin/python my_script.py --kfp-server-host "$API_URL" --namespace
"$NAMESPACE" --token "$OCP_TOKEN"
```

Alternatively, use a prompt with **read -s** to input the token securely at runtime.

3. Optional: If you are running outside the cluster or you use a custom or self-signed certificate, set an environment variable for your trusted certificate bundle:

```
export SSL_CA_CERT=/etc/pki/tls/custom-certs/ca-bundle.crt
```

   Adjust the path if your environment uses a different certificate location.

4. In your Python environment, create a KFP client that uses the pipeline server route and OpenShift access token:

```
import os
from kfp.client import Client

api_url = os.environ["API_URL"]
token = os.environ["OCP_TOKEN"]
namespace = os.environ["NAMESPACE"]

# Optional: Use a custom certificate bundle if required
ssl_ca_cert = os.environ.get("SSL_CA_CERT", None)

client_args = {
    "host": api_url,
    "existing_token": token,
    "namespace": namespace,
}

if ssl_ca_cert:
    client_args["ssl_ca_cert"] = ssl_ca_cert

client = Client(**client_args)
```

5. Verify the connection by calling the API. For example, list experiments or pipelines:

```
print(client.list_experiments())
# or
print(client.list_pipelines())
```

Verification

- The Python code runs without authentication errors.
- The command output lists experiments or pipelines that are defined on the pipeline server for the specified project.

Next steps

- Use the KFP SDK to compile and upload pipelines, create pipeline runs, or manage pipeline versions against the authenticated pipeline server.
- If required, integrate this client configuration into your own automation scripts or external applications that orchestrate pipelines on OpenShift AI.

1.2.4. Defining a pipeline by using the Kubernetes API

You can define AI pipelines and pipeline versions by using the Kubernetes API, which stores them as custom resources in the cluster instead of the internal database. This approach makes it easier to use OpenShift GitOps (Argo CD) or similar tools to manage pipelines and pipeline versions, while still allowing you to manage them through the OpenShift AI dashboard, API, and the Kubeflow Pipelines (KFP) Software Development Kit (SDK). You can generate the required manifests by using the Kubeflow Pipelines SDK; see *Compiling the pipeline YAML with the Kubeflow Pipelines SDK* or *Compiling Kubernetes-native manifests with the Kubeflow Pipelines SDK*.

NOTE

If your pipeline server is already configured to use Kubernetes API storage, you can still use the OpenShift AI dashboard and REST API to view pipeline details, run pipelines, and create schedules. In this mode, the Kubernetes API acts as the storage backend, so your existing tools continue to work as expected.

Prerequisites

- You have OpenShift AI administrator privileges or you are the project owner.
- You have a project with a running pipeline server.
- You have installed the OpenShift CLI (**oc**) as described in the appropriate documentation for your cluster:
  - Installing the OpenShift CLI for OpenShift Container Platform
  - Installing the OpenShift CLI for Red Hat OpenShift Service on AWS
- If you plan to create a **PipelineVersion** custom resource, you have either:
  - Compiled your Python pipeline to IR YAML by using the KFP SDK. See *Compiling the pipeline YAML with the Kubeflow Pipelines SDK*.
  - Compiled Kubernetes-native manifests by using the KFP SDK. See *Compiling Kubernetes-native manifests with the Kubeflow Pipelines SDK*.

Procedure

1. In a terminal window, log in to your OpenShift cluster by using the OpenShift CLI (**oc**):

   ```
   $ oc login -u <user_name>
   ```

   When prompted, enter the OpenShift server URL, connection type, and your password.

2. To configure the pipeline server to use Kubernetes API storage instead of the default **database** option, set the **spec.apiServer.pipelineStore** field to **kubernetes** in your project's **DataSciencePipelinesApplication** (DSPA) custom resource.
   In the following command, replace *<dspa_name>* with the name of your DSPA custom resource, and replace *<namespace>* with the name of your project:

   ```
   $ oc patch dspa <dspa_name> -n <namespace> \
     --type=merge \
     -p {"spec": {"apiServer": {"pipelineStore": "kubernetes"}}}
   ```

WARNING

When you switch the pipeline server from database storage to Kubernetes API storage, existing pipelines that were stored in the internal database are no longer visible in the OpenShift AI dashboard or REST API. To view or manage those pipelines again, change the **spec.apiServer.pipelineStore** field back to **database**.

3. Define a **Pipeline** custom resource in a YAML file with the following contents:

   Example pipeline definition

   ```
   apiVersion: pipelines.kubeflow.org/v2beta1
   kind: Pipeline
   metadata:
     name: <name>
     namespace: <namespace>
   spec:
     displayName: <displayName>
   ```

   - **name**: The immutable Kubernetes resource name of your pipeline.
   - **namespace**: The name of your project.
   - **displayName**: The user-friendly display name of your pipeline, which is shown in the dashboard and REST API.

4. Apply the pipeline definition to create the **Pipeline** custom resource in your cluster.
   In the following command, replace *<pipeline_yaml_file>* with the name of your YAML file:

   Example command

   ```
   $ oc apply -f <pipeline_yaml_file>.yaml
   ```

5. Alternatively, if you compiled Kubernetes-native manifests with the KFP SDK, you can apply the generated file directly without manually creating separate YAML files:

   ```
   $ oc apply -f <output_file>.yaml
   ```

   The generated file includes both **Pipeline** and **PipelineVersion** resources. You can skip the following manual definition steps and proceed to the verification step.

6. Define a **PipelineVersion** custom resource in a YAML file with the following contents:

   Example pipeline version definition

   ```
   apiVersion: pipelines.kubeflow.org/v2beta1
   kind: PipelineVersion
   metadata:
   ```

```
  name: <name>
  namespace: <namespace>
spec:
  pipelineName: <pipelineName>
  displayName: <displayName>
  description: This is the first version of the pipeline.
  pipelineSpec:
      # ... YAML generated by compiling Python pipeline with KFP SDK ...
```

- **name**: The name of your pipeline version.
- **namespace**: The name of your project.
- **pipelineName**: The immutable Kubernetes resource name of your pipeline. This value must match the **metadata.name** value in the **Pipeline** custom resource.
- **displayName**: The user-friendly display name of your pipeline version, which is shown in the dashboard and REST API.
- **pipelineSpec**: The YAML content that you generated by using the Kubeflow Pipelines (KFP) SDK.

7. Apply the pipeline version definition to create the **PipelineVersion** custom resource in your cluster.
   In the following command, replace *<pipeline_version_yaml_file>* with the name of your YAML file:

   #### Example command

   ```
   $ oc apply -f <pipeline_version_yaml_file>.yaml
   ```

   After creating the pipeline version, the system automatically applies the following labels to the pipeline version for easier filtering:

   #### Example automatic labels

   ```
   pipelines.kubeflow.org/pipeline-id: <metadata.uid of the pipeline>
   pipelines.kubeflow.org/pipeline: <pipeline name>
   ```

Verification

1. Check that the **Pipeline** custom resource was successfully created:

   ```
   $ oc get pipeline <pipeline_name> -n <namespace>
   ```

2. Check that the **PipelineVersion** custom resource was successfully created:

   ```
   $ oc get pipelineversion <pipeline_version_name> -n <namespace>
   ```

1.2.5. Migrating pipelines from database to Kubernetes API storage

You can migrate existing pipelines and pipeline versions from the internal database to Kubernetes custom resources. This makes it easier to use OpenShift GitOps (Argo CD) or similar tools to manage pipelines and pipeline versions, while still allowing you to manage them through the OpenShift AI

dashboard, API, and the Kubeflow Pipelines (KFP) Software Development Kit (SDK).

This procedure uses a community-supported Kubeflow Pipelines migration script to export pipelines from the AI Pipelines API and generate corresponding **Pipeline** and **PipelineVersion** custom resources for import into your cluster.

IMPORTANT

The migration script in this procedure is maintained by the Kubeflow Pipelines community and is not supported by Red Hat. Before you use the script, review the repository and validate it in a non-production environment.

WARNING

The pipeline and pipeline version IDs change during migration, so existing pipeline runs do not map to the migrated pipeline version. The original ID is stored in the **pipelines.kubeflow.org/original-id** label.

Prerequisites

- You have OpenShift AI administrator privileges or you are the project owner.
- You have a project with a running pipeline server.
- The pipeline server is configured with **spec.apiServer.pipelineStore: database**.
- You have Python 3.11 installed in your local environment.
- You have installed the OpenShift CLI (**oc**) as described in the appropriate documentation for your cluster:
  - Installing the OpenShift CLI for OpenShift Container Platform
  - Installing the OpenShift CLI for Red Hat OpenShift Service on AWS

Procedure

1. In a terminal window, log in to your OpenShift cluster by using the OpenShift CLI (**oc**):

   ```
   $ oc login -u <user_name>
   ```

   When prompted, enter the OpenShift server URL, connection type, and your password.

2. Set environment variables for your project and get the pipeline API route.
   In the **export** command, replace *<namespace>* with the name of your project:

   ```
   echo "Setting the prerequisite variables"
   export NAMESPACE=<namespace>
   export DSPA_NAME=$(oc -n $NAMESPACE get dspa -o
   ```

```
jsonpath={.items[0].metadata.name})
export API_URL="https://$(oc -n $NAMESPACE get route "ds-pipeline-$DSPA_NAME" -o
jsonpath={.spec.host})"
```

3. Create a Python virtual environment and install the required dependencies.

```
echo "Set up the Python prerequisites"
python3.11 -m venv .venv
./.venv/bin/pip install kfp requests PyYAML
```

4. Download and run the Kubeflow Pipelines community migration script.
   The script connects to the AI Pipelines API, exports all pipelines and versions from the specified project, and generates one YAML file per pipeline in a local **kfp-exported-pipelines**/ directory. Each file includes a **Pipeline** resource followed by all associated **PipelineVersion** resources.

   a. Run the following command:

```
curl -L https://raw.githubusercontent.com/kubeflow/pipelines/refs/heads/master/tools/k8s-
native/migration.py -o migration.py
./.venv/bin/python migration.py --skip-tls-verify --kfp-server-host $API_URL --namespace
$NAMESPACE --token "$(oc whoami --show-token)"
```

NOTE

The **--skip-tls-verify** option disables certificate validation and should be used only in development environments or when connecting to a server with a self-signed certificate. In production environments, provide a valid certificate bundle instead.

Additionally, passing the access token directly on the command line might expose it in shell history or process lists. To reduce this risk, store the token in an environment variable and reference it in your command:

```
export KFP_TOKEN=$(oc whoami --show-token)
./.venv/bin/python migration.py --kfp-server-host $API_URL --namespace
$NAMESPACE --token "$KFP_TOKEN"
```

Alternatively, use a prompt with **read -s** to input the token securely at runtime.

   b. Optional: For more information about the script, run the following command:

```
./.venv/bin/python migration.py --help
```

   c. If you plan to create new or updated **PipelineVersion** custom resources after migration, you can compile your pipeline code by using the Kubeflow Pipelines SDK. For more information, see *Compiling the pipeline YAML with the Kubeflow Pipelines SDK* and *Compiling Kubernetes-native manifests with the Kubeflow Pipelines SDK*.

5. Apply the exported Kubernetes custom resources to your cluster.

```
oc apply -f ./kfp-exported-pipelines
```

6. Change the pipeline server to use Kubernetes API storage.

```
oc -n "$NAMESPACE" patch dspa "$DSPA_NAME" --type=merge -p {"spec":{"apiServer":
{"pipelineStore":"kubernetes"}}}
```

NOTE

To view pipelines that were stored in the internal database and not migrated, you can temporarily change the pipeline server back to **database** storage.

```
oc -n $NAMESPACE patch dspa $DSPA_NAME --type=merge -p {"spec":
{"apiServer":{"pipelineStore":"database"}}}
```

7. Repeat this procedure for each additional project that you want to migrate, changing **NAMESPACE** to the appropriate project name.

8. Optional: Clean up the local environment.

```
rm -rf .venv migration.py
```

Verification

1. Check that the **Pipeline** and **PipelineVersion** custom resources were created in your project:

```
$ oc -n <namespace> get pipelines.pipelines.kubeflow.org
$ oc -n <namespace> get pipelineversions.pipelines.kubeflow.org
```

2. Verify that the pipeline server is using Kubernetes API storage:

```
$ oc -n <namespace> get dspa <dspa_name> -o jsonpath={.spec.apiServer.pipelineStore}
{"\n"}
```

The command should return **kubernetes**.

Additional resources

- Kubeflow Pipelines community migration script source
- Script README and usage details

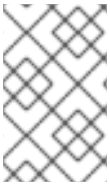
1.3. IMPORTING A PIPELINE

To help you begin working with AI pipelines in OpenShift AI, you can import a YAML file containing your pipeline's code to an active pipeline server, or you can import the YAML file from a URL. This file contains a Kubeflow pipeline compiled by using the Kubeflow compiler. After you have imported the pipeline to a pipeline server, you can execute the pipeline by creating a pipeline run.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.

- You have compiled your pipeline with the Kubeflow compiler and you have access to the resulting YAML file.
- If you are uploading your pipeline from a URL, the URL is publicly accessible.



NOTE

If your pipeline is defined in Python code instead of a YAML file, compile it first by using the KFP SDK. For more information, see *Compiling the pipeline YAML with the Kubeflow Pipelines SDK*.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definition** page, from the **Project** drop-down list, select the project that you want to import a pipeline to.
3. Click **Import pipeline**.
4. In the **Import pipeline** dialog, enter the details for the pipeline that you want to import.
   a. In the **Pipeline name** field, enter a name for the pipeline that you want to import.
   b. In the **Pipeline description** field, enter a description for the pipeline that want to import.
   c. Select where you want to import your pipeline from by performing one of the following actions:
      - Select **Upload a file** to upload your pipeline from your local machine's file system. Import your pipeline by clicking **Upload**, or by dragging and dropping a file.
      - Select **Import by url** to upload your pipeline from a URL, and then enter the URL into the text box.
   d. Click **Import pipeline**.

Verification

- The pipeline that you imported is displayed on the **Pipeline definitions** page and on the **Pipelines** tab on the project details page.

1.4. DELETING A PIPELINE

If you no longer require access to your AI pipeline on the dashboard, you can delete it so that it does not appear on the **Pipeline definitions** page. .Prerequisites * You have logged in to Red Hat OpenShift AI. * There are active pipelines available on the **Pipeline definitions** page. * The pipeline that you want to delete does not contain any pipeline versions. * The pipeline that you want to delete does not contain any pipeline versions. For more information, see Deleting a pipeline version.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipeline that you want to delete.

3. Click the action menu (⋮) beside the pipeline that you want to delete, and then click **Delete pipeline**.
4. In the **Delete pipeline** dialog, enter the pipeline name in the text field to confirm that you intend to delete it.
5. Click **Delete pipeline**.

Verification

- The AI pipeline that you deleted is no longer displayed on the **Pipeline definitions** page.

1.5. DELETING A PIPELINE SERVER

After you have finished running your AI pipelines, you can delete the pipeline server. Deleting a pipeline server automatically deletes all of its associated pipelines, pipeline versions, and runs. If your pipeline data is stored in a database, the database is also deleted along with its meta-data. In addition, after deleting a pipeline server, you cannot create new pipelines or pipeline runs until you create another pipeline server.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipeline server that you want to delete.
3. From the **Pipeline server actions** list, select **Delete pipeline server**.
4. In the **Delete pipeline server** dialog, enter the name of the pipeline server in the text field to confirm that you intend to delete it.
5. Click **Delete**.

Verification

- Pipelines previously assigned to the deleted pipeline server no longer appear on the **Pipeline definitions** page for the relevant project.
- Pipeline runs previously assigned to the deleted pipeline server no longer appear on the **Runs** page for the relevant project.

1.6. VIEWING THE DETAILS OF A PIPELINE SERVER

You can view the details of pipeline servers configured in OpenShift AI, such as the pipeline's connection details and where its data is stored.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that contains an active and available pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipeline server that you want to view.
3. From the **Pipeline server actions** list, select **Manage pipeline server configuration**.

Verification

- You can view the pipeline server details in the **Manage pipeline server** dialog.

1.7. VIEWING EXISTING PIPELINES

You can view the details of pipelines that you have imported to Red Hat OpenShift AI, such as the pipeline's last run, when it was created, the pipeline's executed runs, and details of any associated pipeline versions.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- Existing pipelines are available.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipelines that you want to view.
3. Optional: Click **Expand** ( ) on the row of a pipeline to view its pipeline versions.

Verification

- A list of pipelines is displayed on the **Pipeline definitions** page.

1.8. OVERVIEW OF PIPELINE VERSIONS

You can manage incremental changes to pipelines in OpenShift AI by using versioning. This allows you to develop and deploy pipelines iteratively, preserving a record of your changes. You can track and manage your changes on the OpenShift AI dashboard, allowing you to schedule and execute runs against all available versions of your pipeline.

1.9. UPLOADING A PIPELINE VERSION

You can upload a YAML file to an active pipeline server that contains the latest version of your pipeline, or you can upload the YAML file from a URL. The YAML file must consist of a Kubeflow pipeline compiled by using the Kubeflow compiler. After you upload a pipeline version to a pipeline server, you can execute it by creating a pipeline run.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have a pipeline version available and ready to upload.
- If you are uploading your pipeline version from a URL, the URL is publicly accessible.
- If your pipeline version is based on Python code, compile it to YAML before uploading. For more information, see *Compiling the pipeline YAML with the Kubeflow Pipelines SDK* .

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that you want to upload a pipeline version to.
3. Click the **Import pipeline** drop-down list, and then select **Upload new version**.
4. In the **Upload new version** dialog, enter the details for the pipeline version that you are uploading.
    a. From the **Pipeline** list, select the pipeline that you want to upload your pipeline version to.
    b. In the **Pipeline version name** field, confirm the name for the pipeline version, and change it if necessary.
    c. In the **Pipeline version description** field, enter a description for the pipeline version.
    d. Select where you want to upload your pipeline version from by performing one of the following actions:
        - Select **Upload a file** to upload your pipeline version from your local machine's file system. Import your pipeline version by clicking **Upload**, or by dragging and dropping a file.
        - Select **Import by url** to upload your pipeline version from a URL, and then enter the URL into the text box.
    e. Click **Upload**.

Verification

- The pipeline version that you uploaded is displayed on the **Pipeline definitions** page. Click **Expand** ( ) on the row containing the pipeline to view its versions.

- The **Version** column on the row containing the pipeline version that you uploaded on the **Pipeline definitions** page increments by one.

1.10. DELETING A PIPELINE VERSION

You can delete specific versions of a pipeline when you no longer require them. Deleting a default pipeline version automatically changes the default pipeline version to the next most recent version. If no pipeline versions exist, the pipeline persists without a default version.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**. The **Pipeline definitions** page opens.
2. Delete the pipeline versions that you no longer require:
   - To delete a single pipeline version:
     a. From the **Project** list, select the project that contains a version of a pipeline that you want to delete.
     b. On the row containing the pipeline, click **Expand** ( ).
     c. Click the action menu (⋮) beside the version that you want to delete, and then click **Delete pipeline version**.
     The **Delete pipeline version** dialog opens.
     d. Enter the name of the pipeline version in the text field to confirm that you intend to delete it.
     e. Click **Delete**.
   - To delete multiple pipeline versions:
     a. On the row containing each pipeline version that you want to delete, select the checkbox.
     b. Click the action menu (⋮) next to the **Import pipeline** drop-down list, and then select **Delete** from the list.

Verification

- The pipeline version that you deleted is no longer displayed on the **Pipeline definitions** page or on the **Pipelines** tab for the project.

1.11. VIEWING THE DETAILS OF A PIPELINE VERSION

You can view the details of a pipeline version that you have uploaded to Red Hat OpenShift AI, such as its graph and YAML code.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have a pipeline available on an active and available pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipeline versions that you want to view details for.
3. Click the pipeline name to view further details of its most recent version. The pipeline version details page opens, displaying the **Graph**, **Summary**, and **Pipeline spec** tabs.

   Alternatively, click **Expand** ( ) on the row containing the pipeline that you want to view versions for, and then click the pipeline version that you want to view the details of. The pipeline version details page opens, displaying the **Graph**, **Summary**, and **Pipeline spec** tabs.

Verification

- On the pipeline version details page, you can view the pipeline graph, summary details, and YAML code.

1.12. DOWNLOADING A PIPELINE VERSION

To make further changes to an AI pipeline version that you previously uploaded to OpenShift AI, you can download pipeline version code from the user interface.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have created and imported a pipeline to an active pipeline server that is available to download.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the version that you want to download.
3. Click **Expand** ( ) beside the pipeline that contains the version that you want to download.
4. Click the pipeline version that you want to download.
   The pipeline version details page opens.

5. Click the **Pipeline spec** tab, and then click the **Download** button () to download the YAML file that contains the pipeline version code to your local machine.

Verification

- The pipeline version code downloads to your browser's default directory for downloaded files.

1.13. OVERVIEW OF PIPELINES CACHING

You can use caching within AI pipelines to optimize execution times and improve resource efficiency. Caching reduces redundant task execution by reusing results from previous runs with identical inputs.

Caching is particularly beneficial for iterative tasks, where intermediate steps might not need to be repeated. Understanding caching can help you design more efficient pipelines and save time in model development.

Caching operates by storing the outputs of successfully completed tasks and comparing the inputs of new tasks against previously cached ones. If a match is found, OpenShift AI reuses the cached results instead of re-executing the task, reducing computation time and resource usage.

1.13.1. Caching criteria

For caching to be effective, the following criteria determine if a task can use previously cached results:

- **Input data and parameters** If the input data and parameters for a task are unchanged from a previous run, cached results are eligible for reuse.
- **Task code and configuration**: Changes to the task code or configurations invalidate the cache to ensure that modifications are always reflected.
- **Pipeline environment**: Changes to the pipeline environment, such as dependency versions, also affect caching eligibility to maintain consistency.

1.13.2. Viewing cached steps in the OpenShift AI user interface

Cached steps in pipelines are visually indicated in the user interface (UI):

- Tasks that use cached results display a green icon, helping you quickly identify which steps were cached. The **Status** field in the side panel displays **Cached** for cached tasks.
- The UI also includes information about when the task was previously executed, allowing for easy verification of cache usage.

To check the caching status of specific tasks, navigate to the pipeline details view in the UI. Cached and non-cached tasks are clearly indicated. Cached tasks do not display execution logs because they reuse previously generated outputs and are not re-executed.

1.13.3. Controlling caching in pipelines

Caching is enabled by default in OpenShift AI to improve performance. However, there are instances when disabling caching might be necessary for specific tasks, an entire pipeline, or all pipelines. For example, caching might not be beneficial for tasks that rely on frequently updated data or unique computational needs. In other cases, such as debugging, development, or when deterministic re-execution is required, you might want to disable caching for all pipelines.

CAUTION

Disabling caching at the pipeline or pipeline server level causes all tasks to re-run, potentially increasing compute time and resource usage.

You can control caching for AI pipelines in the following ways:

- **Individual task**: Data scientists can disable caching for specific steps in a pipeline.
- **Pipeline (submit time)**: Data scientists can disable caching when submitting a pipeline run.
- **Pipeline (compile time)**: Data scientists can disable caching when compiling a pipeline.
- **All pipelines (pipeline server)**: You can disable caching for all pipelines in the pipeline server, which overrides all pipeline and task-level caching settings.

1.13.3.1. Disabling caching for individual tasks

To disable caching for a particular task, apply the **set_caching_options** method directly to the task in your pipeline code:

```
task_name.set_caching_options(False)
```

After applying this setting, OpenShift AI runs the task in future pipeline runs, ignoring any cached results.

You can re-enable caching for individual tasks by setting the **set_caching_options** parameter to **True** or by omitting **set_caching_options**.

This setting is ignored if caching is disabled in the pipeline server.

1.13.3.2. Disabling caching for a pipeline at submit time

To disable caching for the entire pipeline during pipeline submission, set the **enable_caching** parameter to **False** in your pipeline code. This setting ensures that no steps are cached during pipeline execution. The **enable_caching** parameter is available only when using the **kfp.client** to submit pipelines or start pipeline runs, such as the **run_pipeline** method.

Example:

```
import kfp
client = kfp.Client()
client.run_pipeline(
    experiment_id=experiment.id,
    pipeline_id=pipeline.id,
    job_name="no-cache-run",
    params={},              # optional
    enable_caching=False,
)
```

This setting is ignored if caching is disabled during pipeline compilation or in the pipeline server.

1.13.3.3. Disabling caching for a pipeline at compile time

To disable caching for the entire pipeline during compilation, set one of the following options in your local environment or workbench:

- Environment variable:

```
export KFP_DISABLE_EXECUTION_CACHING_BY_DEFAULT=true
```

- CLI flag (when using **kfp dsl compile**):

```
kfp dsl compile --disable-execution-caching-by-default
```

These settings are ignored if caching is disabled in the pipeline server.

1.13.3.4. Disabling caching for all pipelines (pipeline server)

To disable caching for all pipelines in the pipeline server and override all pipeline and task-level caching settings, use either of the following methods:

Pipeline server configuration

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Pipeline definitions**.
2. On the **Pipeline definitions** page, from the **Project** drop-down list, select the project that contains the pipeline server that you want to configure.
3. From the **Pipeline server actions** list, select **Manage pipeline server configuration**.
4. In the **Pipeline caching** section, clear the **Allow caching to be configured per pipeline and task** checkbox.
5. Click **Save**.

DataSciencePipelinesApplication (cluster administrator)

In the OpenShift console or CLI, set the **cacheEnabled** field to **false** in the **DataSciencePipelinesApplication** (DSPA) custom resource for the project.
Example:

```
apiVersion: datasciencepipelinesapplications.opendatahub.io/v1
kind: DataSciencePipelinesApplication
metadata:
  name: my-dspa
  namespace: my-namespace
spec:
  apiServer:
    cacheEnabled: false
```

To allow caching to be configured at the pipeline and task level, set the **cacheEnabled** field to **true** in the DSPA custom resource.

After applying this setting, all pipeline and task-level caching settings are ignored.

NOTE

Changing this setting updates the **CACHEENABLED** environment variable in the pipeline server deployment.

Verification

After configuring caching settings, you can verify its behavior by using one of the following methods:

- **Check the UI**: Locate the green icons in the task list to identify cached steps.
- **Test task re-runs**: Disable caching on specific tasks or the pipeline to confirm that steps re-execute as expected.
- **Validate inputs**: Ensure the task inputs, parameters, and runtime settings are unchanged when caching is applied.

NOTE

You can also disable caching for a single node or for your entire pipeline in JupyterLab using Elyra. For more information, see Disabling node caching in Elyra .

Additional resources

- Kubeflow caching documentation: Kubeflow Pipelines - Use Caching
- The **kfp.client** module documentation for the **enable_caching** parameter: KFP SDK API Reference - kfp.client

CHAPTER 2. MANAGING PIPELINE EXPERIMENTS

2.1. OVERVIEW OF PIPELINE EXPERIMENTS

A pipeline experiment is a workspace where you can try different configurations of your pipelines. You can use experiments to organize your runs into logical groups. As a data scientist, you can use OpenShift AI to define, manage, and track pipeline experiments. You can view a record of previously created and archived experiments from the **Experiments** page in the OpenShift AI user interface. Pipeline experiments contain pipeline runs, including recurring runs. This allows you to try different configurations of your pipelines.

When you work with AI pipelines, it is important to monitor and record your pipeline experiments to track the performance of your AI pipelines. You can compare the results of up to 10 pipeline runs at one time, and view available parameter, scalar metric, confusion matrix, and receiver operating characteristic (ROC) curve data for all selected runs.

You can view artifacts for an executed pipeline run from the OpenShift AI dashboard. Pipeline artifacts can help you to evaluate the performance of your pipeline runs and make it easier to understand your pipeline components. Pipeline artifacts can range from plain text data to detailed, interactive data visualizations.

2.2. CREATING A PIPELINE EXPERIMENT

Pipeline experiments are workspaces where you can try different configurations of your pipelines. You can also use experiments to organize your pipeline runs into logical groups. Pipeline experiments contain pipeline runs, including recurring runs.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project to create the pipeline experiment in.
3. Click **Create experiment**.
4. In the **Create experiment** dialog, configure the pipeline experiment:
    a. In the **Experiment name** field, enter a name for the pipeline experiment.
    b. In the **Description** field, enter a description for the pipeline experiment.
    c. Click **Create experiment**.

Verification

- The pipeline experiment that you created is displayed on the **Experiments** tab.

2.3. ARCHIVING A PIPELINE EXPERIMENT

You can archive your pipeline experiments to store and retain records for future reference. If you need to reuse an archived experiment, you can restore it at any time. Deleting pipeline experiments is a separate action and happens only when you explicitly choose to delete them. Unarchived experiments remain stored unless you manually delete them.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- A pipeline experiment is available to archive.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment that you want to archive.
3. Click the action menu (⋮) beside the pipeline experiment that you want to archive, and then click **Archive**.
4. In the **Archiving experiment** dialog, enter the pipeline experiment name in the text field to confirm that you intend to archive it.
5. Click **Archive**.

Verification

- The archived pipeline experiment does not appear on the **Experiments** tab, and instead is displayed on the **Archive** tab on the **Experiments** page for the pipeline experiment.

2.4. DELETING AN ARCHIVED PIPELINE EXPERIMENT

You can delete pipeline experiments from the OpenShift AI experiment archive.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.
- A pipeline experiment is available in the pipeline archive.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.

2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the archived pipeline experiment that you want to delete.
3. Click the **Archive** tab.
4. Click the action menu (⋮) beside the pipeline experiment that you want to delete, and then click **Delete**.
5. In the **Delete experiment?** dialog, enter the pipeline experiment name in the text field to confirm that you intend to delete it.
6. Click **Delete**.

Verification

- The pipeline experiment that you deleted is no longer displayed on the **Archive** tab on the **Experiments** page.

2.5. RESTORING AN ARCHIVED PIPELINE EXPERIMENT

You can restore an archived pipeline experiment to the active state.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- An archived pipeline experiment exists in your project.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the archived pipeline experiment that you want to restore.
3. Click the **Archive** tab.
4. Click the action menu (⋮) beside the pipeline experiment that you want to restore, and then click **Restore**.
5. In the **Restore experiment** dialog, click **Restore**.

Verification

- The restored pipeline experiment is displayed on the **Experiments** tab on the **Experiments** page.

2.6. VIEWING PIPELINE TASK EXECUTIONS

When a pipeline run executes, you can view details of executed tasks in each step in a pipeline run from the OpenShift AI dashboard. A step forms part of a task in a pipeline.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously triggered a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Executions**.
2. On the **Executions** page, from the **Project** drop-down list, select the project that contains the experiment for the pipeline task executions that you want to view.

Verification

- On the **Executions** page, you can view the execution details of each pipeline task execution, such as its name, status, unique ID, and execution type. The execution status indicates whether the pipeline task has successfully executed. For further information about the details of the task execution, click the execution name.

2.7. VIEWING PIPELINE ARTIFACTS

After a pipeline run executes, you can view its pipeline artifacts from the OpenShift AI dashboard. Pipeline artifacts can help you to evaluate the performance of your pipeline runs and make it easier to understand your pipeline components. Pipeline artifacts can range from plain text data to detailed, interactive data visualizations.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously triggered a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Artifacts**.
2. On the **Artifacts** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the pipeline artifacts that you want to view.
3. Click an artifact name in the list to view additional information about the artifact, including its original run, original execution, and properties.
4. To view or download the content of an artifact stored in S3-compatible object storage, click the preview icon, the download icon, or the active artifact URI link.
   Clicking the preview icon or the URI link for content that your browser can display (such as plain text, HTML, or markdown) opens the artifact in a new browser tab. Clicking the download icon or the URI link for content that your browser cannot display (such as a model file) downloads the

artifact. To download an artifact that is displayed in a browser tab, right-click the content and then click **Save as**.

Verification

- On the **Artifacts** page, you can view the details of each pipeline artifact, such as its name, unique ID, type, and URI.

2.8. COMPARING RUNS IN AN EXPERIMENT

You can compare up to 10 pipeline runs in the same experiment at one time, and view available parameter, scalar metric, confusion matrix, and receiver operating characteristic (ROC) curve data for all selected runs.

To compare runs from different experiments or pipelines, or to view every pipeline run in a project, see Comparing runs in different experiments .

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have created at least two pipeline runs.

Procedure

1. In the OpenShift AI dashboard, select **Develop & train → Experiments**.
   The **Experiments** page opens.
2. From the **Project** drop-down list, select the project that contains the runs that you want to compare.
3. On the **Experiments** tab, in the **Experiment** column, click the experiment that you want to compare runs for. To select runs that are not in an experiment, click **Default**. All runs that are created without specifying an experiment will appear in the **Default** group.
   The **Runs** page opens.
4. Select the checkbox next to each run that you want to compare, and then click **Compare runs**. You can compare a maximum of 10 runs at one time.
   The **Compare runs** page opens and displays data for the runs that you selected.
   a. The **Run list** section displays a list of selected runs. You can filter the list by run name, pipeline version, start date, and status.
   b. The **Parameters** section displays parameter information for each selected run. Set the **Hide parameters with no differences** switch to **On** to hide parameters that have the same values.
   c. The **Metrics** section displays scalar metric, confusion matrix, and ROC curve data for all selected runs.
      i. On the **Scalar metrics** tab, set the **Hide parameters with no differences** switch to **On** to hide parameters that have the same values.

ii. On the **ROC curve** tab, in the artifacts list, adjust the ROC curve chart by clearing the checkbox next to artifacts that you want to remove from the chart.

5. To select different runs for comparison, click **Manage runs**.
   The **Manage runs** dialog opens.

   a. From the filter drop-down list, select **Run**, **Pipeline version**, **Created after**, or **Status** to filter the run list by each value.

   b. Clear the checkbox next to each run that you want to remove from your comparison.

   c. Select the checkbox next to each run that you want to add to your comparison.

6. Click **Update**.

Verification

- The **Compare runs** page opens and displays data for the runs that you selected.

2.9. COMPARING RUNS IN DIFFERENT EXPERIMENTS

You can compare up to 10 pipeline runs from any experiment or pipeline in a project, including runs that do not have a corresponding pipeline, and view available parameter, scalar metric, confusion matrix, and receiver operating characteristic (ROC) curve data for all selected runs.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have created at least two pipeline runs.

Procedure

1. In the OpenShift AI dashboard, select **Develop & train → Pipelines → Runs**.
   The **Runs** page opens.

2. From the **Project** drop-down list, select the project that contains the runs that you want to compare.

3. On the **Runs** tab, in the **Run** column, select the checkbox next to each run that you want to compare, and then click **Compare runs**. You can compare a maximum of 10 runs at one time.
   The **Compare runs** page opens and displays data for the runs that you selected.

   a. The **Run list** section displays a list of selected runs. You can filter the list by run name, experiment, pipeline version, start date, and status.

   b. The **Parameters** section displays parameter information for each selected run. Set the **Hide parameters with no differences** switch to **On** to hide parameters that have the same values.

   c. The **Metrics** section displays scalar metric, confusion matrix, and ROC curve data for all selected runs.

i. On the **Scalar metrics** tab, set the **Hide parameters with no differences** switch to **On** to hide parameters that have the same values.

ii. On the **ROC curve** tab, in the artifacts list, adjust the ROC curve chart by clearing the checkbox next to artifacts that you want to remove from the chart.

4. To select different runs for comparison, click **Manage runs**.
   The **Manage runs** page opens.

   a. From the filter drop-down list, select **Run**, **Experiment**, **Pipeline version**, **Created after**, or **Status** to filter the run list by each value.

   b. Clear the checkbox next to each run that you want to remove from your comparison.

   c. Select the checkbox next to each run that you want to add to your comparison.

5. Click **Update**.

Verification

- The **Compare runs** page opens and displays data for the runs that you selected.

CHAPTER 3. MANAGING PIPELINE RUNS

3.1. OVERVIEW OF PIPELINE RUNS

A pipeline run is a single execution of an AI pipeline. As data scientist, you can use OpenShift AI to define, manage, and track executions of a pipeline. To view a record of previously executed, scheduled, and archived runs, you must first select the experiment from the **Develop & train → Experiments** page in the OpenShift AI interface. After selecting the experiment, you can access all of its pipeline runs from the **Runs** page.

You can optimize your use of pipeline runs for portability and repeatability by using pipeline experiments. With experiments, you can logically group pipeline runs and try different configurations of your pipelines. You can also clone your pipeline runs to reproduce and scale them, or archive them when you want to retain a record of their execution, but no longer require them. You can delete archived runs that you no longer want to retain, or you can restore them to their former state.

You can execute a run once, that is, immediately after its creation, or on a recurring basis. Recurring runs consist of a copy of a pipeline with all of its parameter values and a run trigger. A run trigger indicates when a recurring run executes. You can define the following run triggers:

- Periodic: Used for scheduling runs to execute in intervals
- Cron: Used for scheduling runs as a cron job

You can also configure up to 10 instances of the same run to execute concurrently. You can track the progress of a run from the run details page on the OpenShift AI user interface. From here, you can view the graph and output artifacts for the run.

A pipeline run can be in one of the following states:

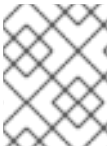
- Scheduled: A pipeline run that is scheduled to execute at least once
- Active: A pipeline run that is executing, or stopped
- Archived: An archived pipeline run

You can use catch up runs to ensure your pipeline runs do not permanently fall behind schedule when paused. For example, if you re-enable a paused recurring run, the run scheduler backfills each missed run interval. If you disable catch up runs, and you have a scheduled run interval ready to execute, the run scheduler only schedules the run execution for the latest run interval. Catch up runs are enabled by default. However, if your pipeline handles backfill internally, Red Hat recommends that you disable catch up runs to avoid duplicate backfill.

After a pipeline run executes, you can view details of its executed tasks on the **Develop & train → Pipelines → Executions** page, along with its artifacts, on the **Develop & train → Pipelines → Artifacts** page. From the **Executions** page, you can view the execution status of each task, which indicates whether it completed successfully. You can also view further information about each executed task by clicking the execution name in the list. From the **Artifacts** page, you can view the details of each pipeline artifact, such as its name, unique ID, type, and URI. Pipeline artifacts can help you to evaluate the performance of your pipeline runs and make it easier to understand your pipeline components. Pipeline artifacts can range from plain text data to detailed, interactive data visualizations.

You can view further information about each artifact, including its original run and original execution, by clicking the artifact name in the list. You can also view or download the content of artifacts stored in S3-compatible object storage by clicking the preview icon, the download icon, or the active artifact URI link.

Clicking the preview icon or the URI link for content that your browser can display (such as plain text, HTML, or markdown) opens the artifact in a new browser tab. Clicking the download icon or the URI link for content that your browser cannot display (such as a model file) downloads the artifact. To download an artifact that is displayed in a browser tab, right-click the content and then click **Save as**.



NOTE

Artifacts that are not stored in S3-compatible object storage are not available to download and do not display an active URI link.

You can review and analyze logs for each step in an active pipeline run. With the log viewer, you can search for specific log messages, view the log for each step, and download the step logs to your local machine.

3.2. STORING DATA WITH PIPELINES

When you run an AI pipeline, OpenShift AI stores the pipeline YAML configuration file and resulting pipeline run artifacts in the **root** directory of your storage bucket. The directories that contain pipeline run artifacts can differ depending on where you executed the pipeline run from. See the following table for further information:

Table 3.1. Pipeline configuration file and artifacts storage locations

Pipeline run source	Pipeline storage directory	Run artifacts storage directory
OpenShift AI dashboard	**/pipelines/<pipeline_version_id>** Example: **/pipelines/1d01c4eb-d2ab-4916-9935-a73a5580f1fb**	**/<pipeline_name>/<pipeline run_id>** Example: **iris-training-pipeline/2g48k8pw-a8ib-4884-9145-h41j7599h3ds**
JupyterLab Elyra extension	**/pipelines/<pipeline_version_id>**	**/<pipeline_name_timestamp>** Example: **/hello-generic-world-0523161704** With the JupyterLab Elyra extension, you can also set an object storage path prefix. Example: **/iris-project/hello-generic-world-0523161704**

If you want to use an existing artifact that was not generated by a task in a pipeline, you can use the kfp.dsl.importer component to import the artifact from its URI. You can only import these artifacts to the S3-compatible object storage bucket that you define in the **Bucket** field in your pipeline server configuration. For more information about the **kfp.dsl.importer** component, see Special Case: Importer Components.

3.3. UNDERSTANDING PIPELINE RUN WORKSPACES

The workspace feature in AI pipelines provides standardized ephemeral shared storage that exists only for the duration of a pipeline run. This storage lets pipeline components exchange large intermediate data without repeatedly uploading to or downloading from object storage, improving performance and reducing object storage costs.

A workspace is a standardized directory structure backed by a Persistent Volume Claim (PVC) that is dynamically created for each pipeline run based on the **DataSciencePipelinesApplication** (DSPA) workspace configuration. When the pipeline definition uses a workspace, the workspace PVC is automatically mounted into each pipeline task pod that requires it. At runtime, the mount path is resolved by using the **dsl.WORKSPACE_PATH_PLACEHOLDER** variable as an input parameter to components.

Pipelines opt into workspace support at compile time. When the pipeline definition includes a workspace configuration, the pipeline compiler annotates the run so that the DSPA automatically creates the workspace PVC and mounts the volume into each task pod that needs it.

3.3.1. Configuring default workspace PVC settings in DSPA

You can configure the default values for workspace Persistent Volume Claims (PVCs) in your **DataSciencePipelinesApplication** (DSPA). This allows you to specify the storage class and access mode for workspace volumes used by pipeline runs. You must provide the workspace storage size when you define the pipeline.

Prerequisites

- You have logged in to OpenShift AI.
- You have the required roles and permissions to edit DSPAs in your project.
- You have configured a pipeline server in your project.

Procedure

1. Log in to the OpenShift web console.
2. Navigate to your project namespace.
3. Click **Search → Resources** and select **DataSciencePipelinesApplication** from the resource list.
4. Select your DSPA instance and click the **YAML** tab to edit the configuration.
5. Add or update the workspace configuration under the **spec.apiServer.workspace** field:

```
apiVersion: datasciencepipelinesapplications.opendatahub.io/v1
kind: DataSciencePipelinesApplication
metadata:
  name: sample-dspa
  namespace: my-namespace
spec:
  dspVersion: v2
  apiServer:
    deploy: true
    workspace:
      volumeClaimTemplateSpec:
        accessModes:
        - ReadWriteOnce
```

```
      storageClassName: standard-csi
  # ... other DSPA configuration
```

- **apiServer.workspace.volumeClaimTemplateSpec** configures the PVC template for workspace volumes.
- **accessModes** specifies the access mode for workspace PVCs. Valid values are **ReadWriteOnce**, **ReadWriteMany**, or **ReadOnlyMany**.
- **storageClassName** specifies the storage class used for workspace PVCs.
- Additional PVC fields supported by Kubernetes, such as **resources** or **persistentVolumeReclaimPolicy**, can also be configured if needed.

6. Click **Save** to apply the changes.

Verification

1. Run the following sample pipeline. It writes a file into the workspace by using **dsl.WORKSPACE_PATH_PLACEHOLDER** and then reads the same file in a later task:

```
from kfp import dsl

@dsl.component
def write_to_workspace(workspace_path: str) -> str:
    """Write a file to the workspace."""
    import os

    file_path = os.path.join(workspace_path, "data", "test_file.txt")
    os.makedirs(os.path.dirname(file_path), exist_ok=True)

    with open(file_path, "w") as f:
        f.write("Hello from workspace!")

    print(f"Wrote file to: {file_path}")
    return file_path

@dsl.component
def read_from_workspace(file_path: str) -> str:
    """Read a file from the workspace using the provided file path."""
    import os

    if os.path.exists(file_path):
        with open(file_path, "r") as f:
            content = f.read()
        print(f"Read content from: {file_path}")
        print(f"Content: {content}")
        assert content == "Hello from workspace!"
        return content

    print(f"File not found at: {file_path}")
    return "File not found"

@dsl.pipeline(
    name="pipeline-with-workspace",
    description="A pipeline that demonstrates workspace functionality",
```

```
    pipeline_config=dsl.PipelineConfig(
        workspace=dsl.WorkspaceConfig(
            size='100Mi',
        ),
    ),
)
def pipeline_with_workspace() -> str:
    """A pipeline using workspace functionality with write and read components."""

    write_task = write_to_workspace(
        workspace_path=dsl.WORKSPACE_PATH_PLACEHOLDER,
    )

    read_task = read_from_workspace(
        file_path=write_task.output,
    )

    return read_task.output
```

2. When the pipeline completes, review the task logs to verify that both write and read operations succeeded.

Additional resources

- Special Case: Importer Components

3.3.2. Adding external artifacts to pipeline run workspaces

When using **dsl.importer** with external artifacts such as Modelcar images stored in an Open Container Initiative (OCI) registry, you cannot download them directly into the workspace. To make the artifact available to subsequent tasks without additional downloads, copy the artifact into the workspace volume in a separate pipeline step.

Prerequisites

- You have configured a pipeline server in your project with workspace support enabled.
- You can reach the external location that stores your artifacts.
- Your pipeline server has the credentials and network access required to pull from that location.

Procedure

To use an external artifact in your pipeline, define a component that copies the artifact to the workspace, and then call this component in your pipeline.

1. Define a component that copies the imported artifact to the workspace:

```
from kfp import dsl

@dsl.component()
def copy_model_to_workspace(input_model: dsl.Input[dsl.Model], workspace_path: str):
    import os
    import shutil
    shutil.copytree(input_model.path, os.path.join(workspace_path, "my-model"))
```

2. Define your pipeline with workspace configuration:

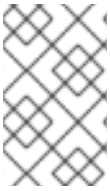
```
@dsl.pipeline(
    name="modelcar-to-workspace",
    pipeline_config=dsl.PipelineConfig(
        workspace=dsl.WorkspaceConfig(size='20Gi'),
    ),
)
def modelcar_to_workspace_pipeline(
    oci_uri: str = "oci://registry.redhat.io/rhelai1/modelcar-granite-3.1-8b-lab-v2.2:latest",
):
    # Import the model from the OCI registry
    model_source = dsl.importer(
        artifact_uri=oci_uri,
        artifact_class=dsl.Model,
    )

    # Copy the model to the workspace
    copy_model_to_workspace(
        input_model=model_source.output,
        workspace_path=dsl.WORKSPACE_PATH_PLACEHOLDER,
    )
```

   - The **pipeline_config** parameter with the workspace size is required to enable workspace support for the pipeline run.

3. Compile and upload your pipeline to OpenShift AI.

Verification

1. Navigate to the **Run details** page in the OpenShift AI dashboard.
2. Verify that the **copy_model_to_workspace** task completes successfully.
3. Check the logs of subsequent tasks to confirm they can access the model files in the workspace (for example, **/kfp-workspace/my-model**).



NOTE

The model files are copied to a subdirectory in the workspace (in this example, **my-model**). Subsequent pipeline tasks can access these files at the workspace mount path resolved at runtime.

Additional resources

- Special Case: Importer Components

3.4. VIEWING ACTIVE PIPELINE RUNS

You can view a list of pipeline runs that were previously executed in a pipeline experiment. From this list, you can view details relating to your pipeline runs, such as the pipeline version that the run belongs to, along with the run status, duration, and execution start time.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously executed a pipeline run that is available.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the active pipeline runs that you want to view.
3. From the list of experiments, click the experiment that contains the active pipeline runs that you want to view.
   The **Runs** page opens.

   After a run has completed its execution, the run status is displayed in the **Status** column of the **Runs** tab, indicating whether the run succeeded or failed.

Verification

- A list of active runs is displayed on the **Runs** tab on the **Runs** page for the pipeline experiment.

3.5. EXECUTING A PIPELINE RUN

By default, a pipeline run executes once immediately after it is created.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment that you want to create a run for.
3. From the list of pipeline experiments, click the experiment that you want to create a run for.
4. Click **Create run**.
5. On the **Create run** page, configure the run:
   a. From the **Experiment** list, select the pipeline experiment that you want to create a run for. Alternatively, to create a new pipeline experiment, click **Create new experiment**, and then complete the relevant fields in the **Create experiment** dialog.
   b. In the **Name** field, enter a name for the run, up to 255 characters.

c. In the **Description** field, enter a description for the run, up to 255 characters.

d. From the **Pipeline** list, select the pipeline that you want to create a run for. Alternatively, to create a new pipeline, click **Create new pipeline**, and then complete the relevant fields in the **Import pipeline** dialog.

e. From the **Pipeline version** list, select the pipeline version to create a run for. Alternatively, to upload a new version, click **Upload new version**, and then complete the relevant fields in the **Upload new version** dialog.

f. Configure the input parameters for the run by selecting the parameters from the list.

g. Click **Create run**.
The details page for the run opens.

Verification

- The pipeline run that you created is displayed on the **Runs** tab on the **Runs** page for the pipeline experiment.

3.6. STOPPING AN ACTIVE PIPELINE RUN

If you no longer require an active pipeline run to continue executing in a pipeline experiment, you can stop the run before its defined end date.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- There is a previously created project available that contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- An active pipeline run is currently executing.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the active run that you want to stop.
3. From the list of pipeline experiments, click the pipeline experiment that contains the run that you want to stop.
4. On the **Runs** tab, click the action menu (⋮) beside the active run that you want to stop, and then click **Stop**.
There might be a short delay while the run stops.

Verification

- The **Failed** status icon () is displayed in the **Status** column of the stopped run.

3.7. DUPLICATING AN ACTIVE PIPELINE RUN

To make it easier to quickly execute pipeline runs with the same configuration in a pipeline experiment, you can duplicate them.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.
- An active run is available to duplicate on the **Active** tab on the **Runs** page.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the pipeline run that you want to duplicate.
3. From the list of pipeline experiments, click the experiment that contains the pipeline run that you want to duplicate.
4. Click the action menu (⋮) beside the relevant active run, and then click **Duplicate**.
5. On the **Duplicate run** page, configure the duplicate run:
    a. From the **Experiment** list, select a pipeline experiment to contain the duplicate run. Alternatively, to create a new pipeline experiment, click **Create new experiment**, and then complete the relevant fields in the **Create experiment** dialog.
    b. In the **Name** field, enter a name for the duplicate run.
    c. In the **Description** field, enter a description for the duplicate run.
    d. From the **Pipeline** list, select a pipeline to contain the duplicate run. Alternatively, to create a new pipeline, click **Create new pipeline**, and then complete the relevant fields in the **Import pipeline** dialog.
    e. From the **Pipeline version** list, select a pipeline version to contain the duplicate run. Alternatively, to upload a new version, click **Upload new version**, and then complete the relevant fields in the **Upload new version** dialog.
    f. In the **Parameters** section, configure input parameters for the duplicate run by selecting parameters from the list.
    g. Click **Create run**.
    The details page for the run opens.

Verification

- The duplicate pipeline run is displayed on the **Runs** tab on the **Runs** page for the pipeline experiment.

3.8. VIEWING SCHEDULED PIPELINE RUNS

You can view a list of pipeline runs that are scheduled for execution in a pipeline experiment. From this list, you can view details relating to your pipeline runs, such as the pipeline version that the run belongs to. You can also view the run status, execution frequency, and schedule.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have scheduled a pipeline run that is available to view.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the scheduled pipeline runs that you want to view.
3. From the list of pipeline experiments, click the experiment that contains the pipeline runs that you want to view.
4. On the **Runs** page, click the **Schedules** tab.
   After a run is scheduled, the **Status** column indicates whether the run is ready or unavailable for execution. To change its execution availability, set the **Status** switch to **On** or **Off**. Alternatively, you can change its execution availability from the details page for the scheduled run by clicking the **Actions** drop-down menu, and then selecting **Enable** or **Disable**.

Verification

- A list of scheduled runs is displayed on the **Schedules** tab on the **Runs** page for the pipeline experiment.

3.9. SCHEDULING A PIPELINE RUN USING A CRON JOB

You can use a cron job to schedule a pipeline run to execute at a specific time. Cron jobs are useful for creating periodic and recurring tasks, and can also schedule individual tasks for a specific time, such as if you want to schedule a run for a low activity period. To successfully execute runs in OpenShift AI, you must use the supported format. See Cron Expression Format for more information.

The following examples show the correct format:

Run occurrence	Cron format
Every five minutes	@every 5m
Every 10 minutes	0 */10 * * * *
Daily at 16:16 UTC	0 16 16 * * *
Daily every quarter of the hour	0 0,15,30,45 * * * *

Run occurrence	Cron format
On Monday and Tuesday at 15:40 UTC	0 40 15 * * MON,TUE

Additional resources

- Cron Expression Format

3.10. SCHEDULING A PIPELINE RUN

To repeatedly run a pipeline, you can create a scheduled pipeline run.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.

Procedure

1. To go to the **Schedules** tab for a run, perform one of the following sets of actions:
   - **To select a run from an experiment**
     a. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
     b. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the run that you want to schedule.
     c. From the list of pipeline experiments, click the experiment that contains the run that you want to schedule.
     d. Click the **Schedules** tab.
   - **To select any run**
     a. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
     b. On the **Runs** page, from the **Project** drop-down list, select the project that contains the run that you want to schedule.
     c. Click the **Schedules** tab.
2. Click **Create schedule**.
3. On the **Create schedule** page, configure the run that you are scheduling:
   a. From the **Experiment** list, select the pipeline experiment that you want to contain the scheduled run. Alternatively, to create a new pipeline experiment, click **Create new experiment**, and then complete the relevant fields in the **Create experiment** dialog.
   b. In the **Name** field, enter a name for the run.

c. In the **Description** field, enter a description for the run.

d. From the **Trigger type** list, select one of the following options:

- Select **Periodic** to specify an execution frequency. In the **Run every** field, enter a number and select an execution frequency from the list.
- Select **Cron** to specify the execution schedule in **cron** format in the **Cron string** field.

  This creates a cron job to execute the run. Click the **Copy** button () to copy the cron job schedule to the clipboard. The field furthest to the left represents seconds. For more information about scheduling tasks using the supported **cron** format, see Cron Expression Format.

e. In the **Maximum concurrent runs** field, specify the number of runs that can execute concurrently, from a range of one to ten.

f. For **Start date**, specify a start date for the run. Select a start date using the calendar, and the start time from the list of times.

g. For **End date**, specify an end date for the run. Select an end date using the calendar, and the end time from the list of times.

h. For **Catch up**, enable or disable catch up runs. You can use catch up runs to ensure your pipeline runs do not permanently fall behind schedule when they are paused. For example, if you re-enable a paused recurring run, the run scheduler backfills each missed run interval.

i. From the **Pipeline** list, select the pipeline that you want to create a run for. Alternatively, to create a new pipeline, click **Create new pipeline**, and then complete the relevant fields in the **Import pipeline** dialog.

j. For **Pipeline version**, select one of the following options:

- Select **Always use the latest pipeline version** so that each recurring run automatically uses the most recent pipeline version.
- Select **Use fixed version**, then select a specific pipeline version for all recurring runs. Alternatively, to upload a new version, click **Upload new version**, and then complete the relevant fields in the **Upload new version** dialog.

k. Configure the input parameters for the run by selecting the parameters from the list.

l. Click **Create schedule**.

Verification

- The pipeline run that you scheduled is displayed on the **Schedules** tab on the **Runs** page for the pipeline experiment.

3.11. DUPLICATING A SCHEDULED PIPELINE RUN

To make it easier to schedule runs to execute as part of your pipeline experiment, you can duplicate existing scheduled runs.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.
- A scheduled run is available to duplicate on the **Schedules** tab on the **Runs** page.

Procedure

1. To go to the **Schedules** tab for a run, perform one of the following sets of actions:
   - **To select a run from an experiment**
     a. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
     b. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the run that you want to duplicate.
     c. From the list of pipeline experiments, click the experiment that contains the run that you want to duplicate.
     d. Click the **Schedules** tab.
   - **To select any run**
     a. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
     b. On the **Runs** page, from the **Project** drop-down list, select the project that contains the run that you want to schedule.
     c. Click the **Schedules** tab.
2. Click the action menu (⋮) beside the run that you want to duplicate, and then click **Duplicate**.
3. On the **Duplicate schedule** page, configure the duplicate run:
   a. From the **Experiment** list, select a pipeline experiment to contain the duplicate run. Alternatively, to create a new pipeline experiment, click **Create new experiment**, and then complete the relevant fields in the **Create experiment** dialog.
   b. In the **Name** field, enter a name for the duplicate run.
   c. In the **Description** field, enter a description for the duplicate run.
   d. From the **Trigger type** list, select one of the following options:
      - Select **Periodic** to specify an execution frequency. In the **Run every** field, enter a number, and select an execution frequency from the list.
      - Select **Cron** to specify the execution schedule in **cron** format in the **Cron string** field.

        This creates a cron job to execute the run. Click the **Copy** button () to copy the cron job schedule to the clipboard. The field furthest to the left represents seconds. For more information about scheduling tasks using the supported **cron** format, see Cron Expression Format.

e. For **Maximum concurrent runs**, specify the number of runs that can execute concurrently, from a range of one to ten.

f. For **Start date**, specify a start date for the duplicate run. Select a start date using the calendar, and the start time from the list of times.

g. For **End date**, specify an end date for the duplicate run. Select an end date using the calendar, and the end time from the list of times.

h. For **Catch up**, enable or disable catch up runs. You can use catch up runs to ensure your pipeline runs do not permanently fall behind schedule when they are paused. For example, if you re-enable a paused recurring run, the run scheduler backfills each missed run interval.

i. From the **Pipeline** list, select the pipeline that you want to create a duplicate run for. Alternatively, to create a new pipeline, click **Create new pipeline**, and then complete the relevant fields in the **Import pipeline** dialog.

j. For **Pipeline version**, select one of the following options:

- Select **Always use the latest pipeline version** so that each recurring run automatically uses the most recent pipeline version.
- Select **Use fixed version**, then select a specific pipeline version for all recurring runs. Alternatively, to upload a new version, click **Upload new version**, and then complete the relevant fields in the **Upload new version** dialog.

k. Configure input parameters for the run by selecting parameters from the list.

l. Click **Schedule run**.

Verification

- The pipeline run that you duplicated is displayed on the **Schedules** tab on the **Runs** page for the pipeline experiment.

3.12. DELETING A SCHEDULED PIPELINE RUN

To discard pipeline runs that you previously scheduled, but no longer require, you can delete them so that they do not appear on the **Schedules** page.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously scheduled a run that is available to delete.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the scheduled pipeline run that you want to delete.

3. From the list of pipeline experiments, click the experiment that contains the scheduled pipeline run that you want to delete.
4. On the **Runs** page, click the **Schedules** tab.
5. Click the action menu (⋮) beside the scheduled pipeline run that you want to delete, and then click **Delete**.
6. In the **Delete schedule** dialog, enter the run name in the text field to confirm that you intend to delete it.
7. Click **Delete**.

Verification

- The run that you deleted is no longer displayed on the **Schedules** tab for the pipeline experiment.

3.13. VIEWING THE DETAILS OF A PIPELINE RUN

To gain a clearer understanding of your pipeline runs, you can view the details of a previously triggered pipeline run, such as its graph, execution details, and run output.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously triggered a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
2. On the **Runs** page, from the **Project** drop-down list, select the project that you want to view the details of a pipeline run for.
3. On the **Runs** page, click the name of the run that you want to view the details of.
   The details page for the run opens.

Verification

- On the run details page, you can view the run graph, execution details, input parameters, step logs, and run output.

3.14. VIEWING ARCHIVED PIPELINE RUNS

You can view a list of pipeline runs that you have archived. You can view details for your archived pipeline runs, such as the pipeline version, run status, duration, and execution start date.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- An archived pipeline run exists.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the archived pipeline runs that you want to view.
3. From the list of pipeline experiments, click the experiment that contains the archived pipeline runs that you want to view.
4. On the **Runs** page, click the **Archive** tab.

Verification

- A list of archived runs is displayed on the **Archive** tab on the **Runs** page for the pipeline experiment.

3.15. ARCHIVING A PIPELINE RUN

You can retain records of your pipeline runs by archiving them. If required, you can restore runs from your archive to reuse, or delete runs that are no longer required.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously executed a pipeline run that is available.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment for the run that you want to archive.
3. From the list of pipeline experiments, click the experiment that contains the pipeline run that you want to archive.
   The **Runs** page opens.
4. On the **Runs** tab, click the action menu (⋮) beside the pipeline run that you want to archive, and then click **Archive**.
5. In the **Archiving run** dialog, enter the run name in the text field to confirm that you intend to archive it.

6. Click **Archive**.

Verification

- The archived run does not appear on the **Runs** tab, and instead is displayed on the **Archive** tab on the **Runs** page for the pipeline experiment.

3.16. RESTORING AN ARCHIVED PIPELINE RUN

You can restore an archived run to the active state.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- An archived run exists in your project.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Experiments**.
2. On the **Experiments** page, from the **Project** drop-down list, select the project that contains the pipeline experiment that you want to restore.
3. From the list of pipeline experiments, click the experiment that contains the archived pipeline run that you want to restore.
4. On the **Runs** page, click the **Archive** tab.
5. Click the action menu (⋮) beside the pipeline run that you want to restore, and then click **Restore**.
6. In the **Restore run?** dialog, click **Restore**.

Verification

- The restored run is displayed on the **Runs** tab on the **Runs** page for the pipeline experiment.

3.17. DELETING AN ARCHIVED PIPELINE RUN

You can delete pipeline runs from the OpenShift AI run archive.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and has a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously archived a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
2. On the **Runs** page, from the **Project** drop-down list, select the project that you want to delete an archived pipeline run from.
3. Click the **Archive** tab.
4. Click the action menu (⋮) beside the pipeline run that you want to delete, and then click **Delete**.
5. In the **Delete run?** dialog, enter the run name in the text field to confirm that you intend to delete it.
6. Click **Delete**.

Verification

- The archived run that you deleted is no longer displayed on the **Archive** tab on the **Runs** page.

3.18. DUPLICATING AN ARCHIVED PIPELINE RUN

To make it easier to reproduce runs with the same configuration as runs in your archive, you can duplicate them.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a configured pipeline server.
- You have imported a pipeline to an active pipeline server.
- An archived run is available to duplicate on the **Archived** tab on the **Runs** page.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
2. On the **Runs** page, from the **Project** drop-down list, select the project that you want to duplicate a pipeline run for.
3. On the **Runs** page, click the **Archive** tab.
4. Click the action menu (⋮) beside the pipeline run that you want to duplicate, and then click **Duplicate**.
5. On the **Duplicate run** page, configure the duplicate run:
    a. From the **Experiment** list, select a pipeline experiment to contain the duplicate run. Alternatively, to create a new pipeline experiment, click **Create new experiment**, and then complete the relevant fields in the **Create experiment** dialog.
    b. In the **Name** field, enter a name for the duplicate run.
    c. In the **Description** field, enter a description for the duplicate run.

d. From the **Pipeline** list, select a pipeline to contain the duplicate run. Alternatively, to create a new pipeline, click **Create new pipeline**, and then complete the relevant fields in the **Import pipeline** dialog.

e. From the **Pipeline version** list, select a pipeline version to contain the duplicate run. Alternatively, to upload a new version, click **Upload new version**, and then complete the relevant fields in the **Upload new version** dialog.

f. In the **Parameters** section, configure input parameters for the duplicate run by selecting parameters from the list.

g. Click **Create run**.
The details page for the run opens.

Verification

- The duplicate pipeline run is displayed on the **Runs** tab on the **Runs** page for the pipeline experiment.

CHAPTER 4. WORKING WITH PIPELINE LOGS

4.1. ABOUT PIPELINE LOGS

You can review and analyze step logs for each step in a triggered pipeline run.

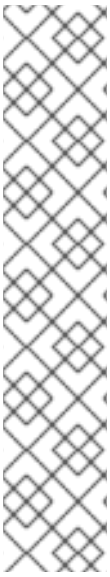
To help you troubleshoot and audit your pipelines, you can review and analyze these step logs by using the log viewer in the OpenShift AI dashboard. From here, you can search for specific log messages, view the log for each step, and download the step logs to your local machine.

If the step log file exceeds its capacity, a warning is displayed above the log viewer stating that the log window displays partial content. Expanding the warning displays further information, such as how the log viewer refreshes every three seconds, and that each step log displays the last 500 lines of log messages received. In addition, you can click **download all step logs** to download all step logs to your local machine.

Each step has a set of container logs. You can view these container logs by selecting a container from the **Steps** list in the log viewer. The **Step-main** container log consists of the log output for the step. The **step-copy-artifact** container log consists of output relating to artifact data sent to s3-compatible storage. If the data transferred between the steps in your pipeline is larger than 3 KB, five container logs are typically available. These logs contain output relating to data transferred between your persistent volume claims (PVCs).

4.2. VIEWING PIPELINE STEP LOGS

To help you troubleshoot and audit your pipelines, you can review and analyze the log of each pipeline step using the log viewer. From here, you can search for specific log messages and download the logs for each step in your pipeline. If the pipeline is running, you can also pause and resume the log from the log viewer.



NOTE

Logs are no longer stored in S3-compatible storage for Python scripts which are running in Elyra pipelines. From OpenShift AI version 2.11, you can view these logs in the pipeline step log viewer.

For this change to take effect, you must use the Elyra runtime images provided in workbench images at version 2024.1 or later.

If you have an older workbench image version, update the **Version selection** field to a compatible workbench image version, for example, **2024.1**, as described in Updating a project workbench.

Updating your workbench image version will clear any existing runtime image selections for your pipeline. After you update your workbench version, open your workbench IDE and update the properties of your pipeline to select a runtime image.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.

- You have previously triggered a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
2. On the **Runs** page, from the **Project** drop-down list, select the project that you want to view pipeline step logs for.
3. On the **Runs** page, click the name of the run that you want to view logs for.
4. On the run details page, on the **Graph** tab, click the pipeline step that you want to view logs for.
5. Click the **Logs** tab.
6. To view the logs of another pipeline step, from the **Steps** list, select the step that you want to view logs for.
7. Analyze the log using the log viewer.
   - To search for a specific log message, enter at least part of the message in the search bar.
   - To view the full log in a separate browser window, click the action menu (⋮) and select **View raw logs**. Alternatively, to expand the size of the log viewer, click the action menu (⋮) and select **Expand**.

Verification

- You can view the logs for each step in your pipeline.

4.3. DOWNLOADING PIPELINE STEP LOGS

Instead of viewing the step logs of a pipeline run using the log viewer on the OpenShift AI dashboard, you can download them for further analysis. You can choose to download the logs belonging to all steps in your pipeline, or you can download the log only for the step log displayed in the log viewer.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have previously created a project that is available and contains a pipeline server.
- You have imported a pipeline to an active pipeline server.
- You have previously triggered a pipeline run.

Procedure

1. From the OpenShift AI dashboard, click **Develop & train → Pipelines → Runs**.
2. On the **Runs** page, from the **Project** drop-down list, select the project that you want to download logs for.
3. On the **Runs** page, click the name of the run that you want to download logs for.

4. On the run details page, on the **Graph** tab, click the pipeline step that you want to download logs for.
5. Click the **Logs** tab.
6. In the log viewer, click the **Download** button ().
    a. Select **Download current stop log** to download the log for the current pipeline step.
    b. Select **Download all step logs** to download the logs for all steps in your pipeline run.

Verification

- The step logs download to your browser's default directory for downloaded files.

CHAPTER 5. WORKING WITH PIPELINES IN JUPYTERLAB

5.1. OVERVIEW OF PIPELINES IN JUPYTERLAB

You can use Elyra to create visual end-to-end pipeline workflows in JupyterLab. Elyra is an extension for JupyterLab that provides you with a Pipeline Editor to create pipeline workflows that can be executed in OpenShift AI.

You can access the Elyra extension within JupyterLab when you create the most recent version of one of the following workbench images:

- Standard Data Science
- PyTorch
- TensorFlow
- TrustyAI
- ROCm-PyTorch
- ROCm-TensorFlow

The Elyra pipeline editor is only available in specific workbench images. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

When you use the Pipeline Editor to visually design your pipelines, minimal coding is required to create and run pipelines. For more information about Elyra, see Elyra Documentation. For more information about the Pipeline Editor, see Visual Pipeline Editor. After you have created your pipeline, you can run it locally in JupyterLab, or remotely using AI pipelines in OpenShift AI.

The pipeline creation process consists of the following tasks:

- Create a project that contains a workbench.
- Create a pipeline server.
- Create a new pipeline in the Pipeline Editor in JupyterLab.
- Develop your pipeline by adding Python notebooks or Python scripts and defining their runtime properties.
- Define execution dependencies.
- Run or export your pipeline.

Before you can run a pipeline in JupyterLab, your pipeline instance must contain a runtime configuration. A runtime configuration defines connectivity information for your pipeline instance and S3-compatible cloud storage.

If you create a workbench as part of a project, a default runtime configuration is created automatically. However, if you create a workbench from the **Start basic workbench** tile in the OpenShift AI dashboard, you must create a runtime configuration before you can run your pipeline in JupyterLab. For more

information about runtime configurations, see Runtime Configuration. As a prerequisite, before you create a workbench, ensure that you have created and configured a pipeline server within the same project as your workbench.

You can use S3-compatible cloud storage to make data available to your notebooks and scripts while they are executed. Your cloud storage must be accessible from the machine in your deployment that runs JupyterLab and from the cluster that hosts AI pipelines. Before you create and run pipelines in JupyterLab, ensure that you have your s3-compatible storage credentials readily available.

Additional resources

- Elyra Documentation
- Visual Pipeline Editor
- Runtime Configuration.

5.2. ACCESSING THE PIPELINE EDITOR

You can use Elyra to create visual end-to-end pipeline workflows in JupyterLab. Elyra is an extension for JupyterLab that provides you with a Pipeline Editor to create pipeline workflows that can execute in OpenShift AI.

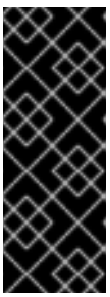
Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project.
- You have created and configured a pipeline server within the project that contains your workbench.

IMPORTANT

To ensure that the runtime configuration is created automatically, you must create the pipeline server before you create the workbench.

- You have created a workbench with a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch), as described in Creating a workbench and selecting an IDE .
- You have started the workbench and opened the JupyterLab interface, as described in Accessing your workbench IDE .



IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

- You have access to S3-compatible storage.

Procedure

1. After you open JupyterLab, confirm that the JupyterLab launcher is automatically displayed.
2. In the **Elyra** section of the JupyterLab launcher, click the **Pipeline Editor** tile.
   The Pipeline Editor opens.

Verification

- You can view the Pipeline Editor in JupyterLab.

5.3. DISABLING NODE CACHING IN ELYRA

You can use the Elyra feature in OpenShift AI to cache components, or "nodes", within your AI pipelines. When a pipeline component runs, Elyra stores its outputs by default. In subsequent runs, if Elyra detects that a particular component has already been executed and its inputs have not changed, it reuses the cached outputs instead of re-running the entire component.

For more information about AI pipelines caching, see Overview of pipelines caching.

If you do not want your component output to be cached, you can disable this feature for a single node or for all of the nodes in your pipeline.

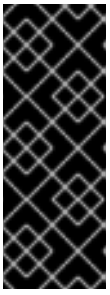
Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project.
- You have created and configured a pipeline server within the project that contains your workbench.

IMPORTANT

To ensure that the runtime configuration is created by default, you must create the pipeline server before you create the workbench.

- You have created a workbench with a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch), as described in Creating a workbench and selecting an IDE .
- You have started the workbench and opened the JupyterLab interface, as described in Accessing your workbench IDE .



IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image that includes the Elyra extension. The Elyra extension is not available in code-server or RStudio IDEs, and it is not included in Minimal Python workbenches. Supported JupyterLab-based images such as Standard Data Science, TensorFlow, PyTorch, TrustyAI, and ROCm variants include the Elyra extension.

- You have access to S3-compatible storage.

- You have created a pipeline in JupyterLab.

Procedure

1. After you open JupyterLab, confirm that the JupyterLab launcher is automatically displayed.
2. Open the pipeline that includes the nodes that you want to modify.
3. Right-click the node that you want to edit, and then click **Open Properties**.
4. To disable caching on a single node, complete the following steps:
    a. Click the **Node Properties** tab in the slide-out menu on the right.
    b. Under **Additional Properties**, click the option bar under **Disable node caching** that is automatically populated with **Use runtime default**.
    c. Select **True**.
5. To disable caching for all nodes on your pipeline, complete the following steps:
    a. Click the **Pipeline Properties** tab in the slide-out menu on the right.
    b. Under **Node Defaults**, click the option bar under **Disable node caching** that is automatically populated with **Use runtime default**.
    c. Select **True**.

Verification

- To verify that caching is disabled for a single node, check that node runs are re-executed in your target runtime environment.
- To verify that caching is disabled for all nodes on your pipeline, check that your entire pipeline runs are re-executed in your target runtime environment.

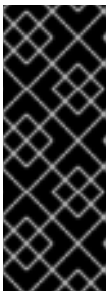
5.4. CREATING A RUNTIME CONFIGURATION

If you create a workbench as part of a project, a default runtime configuration is created automatically. However, if you create a workbench from the **Start basic workbench** tile in the OpenShift AI dashboard, you must create a runtime configuration before you can run your pipeline in JupyterLab. This enables you to specify connectivity information for your pipeline instance and S3-compatible cloud storage.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have access to S3-compatible cloud storage.
- You have created a project that contains a workbench.
- You have created and configured a pipeline server within the project that contains your workbench.

- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).



IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

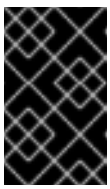
Procedure

1. In the left sidebar of JupyterLab, click **Runtimes** ( ).
2. Click the **Create new runtime configuration** button ( ).
   The **Add new Pipelines runtime configuration** page opens.
3. Complete the relevant fields to define your runtime configuration.
   a. In the **Display Name** field, enter a name for your runtime configuration.
   b. Optional: In the **Description** field, enter a description to define your runtime configuration.
   c. Optional: In the **Tags** field, click **Add Tag** to define a category for your pipeline instance. Enter a name for the tag and press Enter.
   d. Define the credentials of your pipeline:
      i. In the **API endpoint** field, enter the API endpoint of your pipeline. Do not specify the pipelines namespace in this field.

IMPORTANT

With Elyra, you can now use a service-based URL instead of the route-based URL by including the port number. Using the service-based URL allows your pipeline to access the service directly.

      ii. In the **Public API endpoint** field, enter the public API endpoint of your pipeline.



IMPORTANT

You can obtain the pipeline API endpoint from the **Develop & train → Pipelines → Runs** page in the dashboard. Copy the relevant endpoint and enter it in the **Public API endpoint** field.

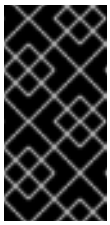
      iii. Optional: In the **User namespace** field, enter the relevant user namespace to run pipelines.
      iv. From the **Authentication Type** list, select the authentication type required to authenticate your pipeline.

IMPORTANT

If you created a workbench directly from the **Start basic workbench** tile on the dashboard, select **EXISTING_BEARER_TOKEN** from the **Authentication Type** list.

v. In the **API endpoint username** field, enter the user name required for the authentication type.

vi. In the **API endpoint password or token**, enter the password or token required for the authentication type.



IMPORTANT

To obtain the pipeline API endpoint token, in the upper-right corner of the OpenShift web console, click your user name and select **Copy login command**. After you have logged in, click **Display token** and copy the value of **--token=** from the **Log in with this token** command.

e. Define the connectivity information of your S3-compatible storage:

i. In the **Cloud Object Storage Endpoint** field, enter the endpoint of your S3-compatible storage. For more information about Amazon s3 endpoints, see Amazon Simple Storage Service endpoints and quotas.

ii. Optional: In the **Public Cloud Object Storage Endpoint** field, enter the URL of your S3-compatible storage.

iii. In the **Cloud Object Storage Bucket Name** field, enter the name of the bucket where your pipeline artifacts are stored. If the bucket name does not exist, it is created automatically.

iv. From the **Cloud Object Storage Authentication Type** list, select the authentication type required to access to your S3-compatible cloud storage. If you use AWS S3 buckets, select **KUBERNETES_SECRET** from the list.

v. In the **Cloud Object Storage Credentials Secret** field, enter the secret that contains the storage user name and password. This secret is defined in the relevant user namespace, if applicable. In addition, it must be stored on the cluster that hosts your pipeline runtime.

vi. In the **Cloud Object Storage Username** field, enter the user name to connect to your S3-compatible cloud storage, if applicable. If you use AWS S3 buckets, enter your AWS Secret Access Key ID.

vii. In the **Cloud Object Storage Password** field, enter the password to connect to your S3-compatible cloud storage, if applicable. If you use AWS S3 buckets, enter your AWS Secret Access Key.

f. Click **Save & Close**.

Verification

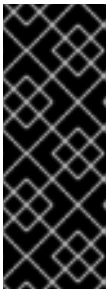
- The runtime configuration that you created is displayed on the **Runtimes** tab ( ) in the left sidebar of JupyterLab.

5.5. UPDATING A RUNTIME CONFIGURATION

To ensure that your runtime configuration is accurate and updated, you can change the settings of an existing runtime configuration.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have access to S3-compatible storage.
- You have created a project that contains a workbench.
- You have created and configured a pipeline server within the project that contains your workbench.
- A previously created runtime configuration is available in the JupyterLab interface.
- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).



IMPORTANT

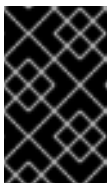
The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

Procedure

1. In the left sidebar of JupyterLab, click **Runtimes** ( ).
2. Hover the cursor over the runtime configuration that you want to update and click the **Edit** button ( ).
   The **Pipelines runtime configuration** page opens.
3. Fill in the relevant fields to update your runtime configuration.
   a. In the **Display Name** field, update name for your runtime configuration, if applicable.
   b. Optional: In the **Description** field, update the description of your runtime configuration, if applicable.
   c. Optional: In the **Tags** field, click **Add Tag** to define a category for your pipeline instance. Enter a name for the tag and press Enter.

d. Define the credentials of your pipeline:

i. In the **API Endpoint** field, update the API endpoint of your AI pipeline, if applicable. Do not specify the pipelines namespace in this field.



IMPORTANT

With Elyra, you can now use a service-based URL instead of the route-based URL by including the port number. Using the service-based URL allows your AI pipeline to access the service directly.

ii. In the **Public API endpoint** field, update the API endpoint of your AI pipeline, if applicable.

iii. Optional: In the **User namespace** field, update the relevant user namespace to run pipelines, if applicable.

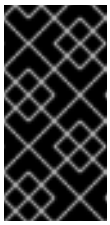
iv. From the **Authentication Type** list, select a new authentication type required to authenticate your pipeline, if applicable.

IMPORTANT

If you created a workbench directly from the **Start basic workbench** tile on the dashboard, select **EXISTING_BEARER_TOKEN** from the **Authentication Type** list.

v. In the **API endpoint username** field, update the user name required for the authentication type, if applicable.

vi. In the **API endpoint password or token**, update the password or token required for the authentication type, if applicable.



IMPORTANT

To obtain the pipelines API endpoint token, in the upper-right corner of the OpenShift web console, click your user name and select **Copy login command**. After you have logged in, click **Display token** and copy the value of **--token=** from the **Log in with this token** command.

e. Define the connectivity information of your S3-compatible storage:

i. In the **Cloud Object Storage Endpoint** field, update the endpoint of your S3-compatible storage, if applicable. For more information about Amazon s3 endpoints, see Amazon Simple Storage Service endpoints and quotas .

ii. Optional: In the **Public Cloud Object Storage Endpoint** field, update the URL of your S3-compatible storage, if applicable.

iii. In the **Cloud Object Storage Bucket Name** field, update the name of the bucket where your pipeline artifacts are stored, if applicable. If the bucket name does not exist, it is created automatically.

iv. From the **Cloud Object Storage Authentication Type** list, update the authentication type required to access to your S3-compatible cloud storage, if applicable. If you use AWS S3 buckets, you must select **USER_CREDENTIALS** from the list.

v. Optional: In the **Cloud Object Storage Credentials Secret** field, update the secret that contains the storage user name and password, if applicable. This secret is defined in the relevant user namespace. You must save the secret on the cluster that hosts your pipeline runtime.

vi. Optional: In the **Cloud Object Storage Username** field, update the user name to connect to your S3-compatible cloud storage, if applicable. If you use AWS S3 buckets, update your AWS Secret Access Key ID.

vii. Optional: In the **Cloud Object Storage Password** field, update the password to connect to your S3-compatible cloud storage, if applicable. If you use AWS S3 buckets, update your AWS Secret Access Key.

f. Click **Save & Close**.

Verification

- The runtime configuration that you updated is shown on the **Runtimes** tab ( ) in the left sidebar of JupyterLab.

5.6. DELETING A RUNTIME CONFIGURATION

After you have finished using your runtime configuration, you can delete it from the JupyterLab interface. After deleting a runtime configuration, you cannot run pipelines in JupyterLab until you create another runtime configuration.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project that contains a workbench.
- You have created and configured a pipeline server within the project that contains your workbench.
- A previously created runtime configuration is visible in the JupyterLab interface.
- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).

IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

Procedure

1. In the left sidebar of JupyterLab, click **Runtimes** ( ).
2. Hover the cursor over the runtime configuration that you want to delete and click the **Delete Item** button ( ).
   A dialog box opens prompting you to confirm the deletion of your runtime configuration.
3. Click **OK**.

Verification

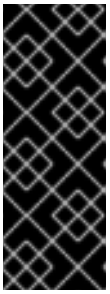
- The runtime configuration that you deleted is no longer shown on the **Runtimes** tab ( ) in the left sidebar of JupyterLab.

5.7. DUPLICATING A RUNTIME CONFIGURATION

To prevent you from re-creating runtime configurations with similar values in their entirety, you can duplicate an existing runtime configuration in the JupyterLab interface.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project that contains a workbench.
- You have created and configured a pipeline server within the project that contains your workbench.
- A previously created runtime configuration is visible in the JupyterLab interface.
- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).



IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

Procedure

1. In the left sidebar of JupyterLab, click **Runtimes** ( ).
2. Hover the cursor over the runtime configuration that you want to duplicate and click the **Duplicate** button ( ).

Verification

- The runtime configuration that you duplicated is shown on the **Runtimes** tab ( ) in the left sidebar of JupyterLab.

5.8. RUNNING A PIPELINE IN JUPYTERLAB

You can run pipelines that you have created in JupyterLab from the Pipeline Editor user interface. Before you can run a pipeline, you must create a project and a pipeline server. After you create a pipeline server, you must create a workbench within the same project as your pipeline server. Your pipeline instance in JupyterLab must contain a runtime configuration. If you create a workbench as part of a project, a default runtime configuration is created automatically. However, if you create a workbench from the **Start basic workbench** tile in the OpenShift AI dashboard, you must create a runtime configuration before you can run your pipeline in JupyterLab. A runtime configuration defines connectivity information for your pipeline instance and S3-compatible cloud storage.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have access to S3-compatible storage.
- You have created a pipeline in JupyterLab.
- You have opened your pipeline in the Pipeline Editor in JupyterLab.
- Your pipeline instance contains a runtime configuration.
- You have created and configured a pipeline server within the project that contains your workbench.
- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).

IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

Procedure

1. In the Pipeline Editor user interface, click **Run Pipeline** ( ).
   The **Run Pipeline** dialog opens. The **Pipeline Name** field is automatically populated with the pipeline file name.

NOTE

After you run your pipeline, a pipeline experiment containing your pipeline run is automatically created on the **Develop & train → Experiments** page in the OpenShift AI dashboard. The experiment name matches the name that you assigned to the pipeline.

2. Define the settings for your pipeline run.
    a. From the **Runtime Configuration** list, select the relevant runtime configuration to run your pipeline.
    b. Optional: Configure your pipeline parameters, if applicable. If your pipeline contains nodes that reference pipeline parameters, you can change the default parameter values. If a parameter is required and has no default value, you must enter a value.
3. Click **OK**.

Verification

- You can view the details of your pipeline run on the **Develop & train → Experiments** page in the OpenShift AI dashboard.
- You can view the output artifacts of your pipeline run. The artifacts are stored in your designated object storage bucket.

5.9. EXPORTING A PIPELINE IN JUPYTERLAB

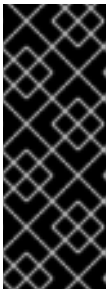
You can export pipelines that you have created in JupyterLab. When you export a pipeline, the pipeline is prepared for later execution, but is not uploaded or executed immediately. During the export process, any package dependencies are uploaded to S3-compatible storage. Also, pipeline code is generated for the target runtime.

Before you can export a pipeline, you must create a project and a pipeline server. After you create a pipeline server, you must create a workbench within the same project as your pipeline server. In addition, your pipeline instance in JupyterLab must contain a runtime configuration. If you create a workbench as part of a project, a default runtime configuration is created automatically. However, if you create a workbench from the **Start basic workbench** tile in the OpenShift AI dashboard, you must create a runtime configuration before you can export your pipeline in JupyterLab. A runtime configuration defines connectivity information for your pipeline instance and S3-compatible cloud storage.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have created a project that contains a workbench.
- You have created and configured a pipeline server within the project that contains your workbench.
- You have access to S3-compatible storage.
- You have a created a pipeline in JupyterLab.
- You have opened your pipeline in the Pipeline Editor in JupyterLab.

- Your pipeline instance contains a runtime configuration.
- You have created and launched a workbench from a workbench image that contains the Elyra extension (Standard Data Science, TensorFlow, TrustyAI, ROCm-PyTorch, ROCm-TensorFlow, or PyTorch).



IMPORTANT

The Elyra pipeline editor is available in specific workbench images only. To use Elyra, the workbench must be based on a JupyterLab image. The Elyra extension is not available in code-server or RStudio IDEs. The workbench must also be derived from the Standard Data Science image. It is not available in Minimal Python or CUDA-based workbenches. All other supported JupyterLab-based workbench images have access to the Elyra extension.

Procedure

1. In the Pipeline Editor user interface, click **Export Pipeline** ().
   The **Export Pipeline** dialog opens. The **Pipeline Name** field is automatically populated with the pipeline file name.
2. Define the settings to export your pipeline.
   a. From the **Runtime Configuration** list, select the relevant runtime configuration to export your pipeline.
   b. From the **Export Pipeline as** select an appropriate file format
   c. In the **Export Filename** field, enter a file name for the exported pipeline.
   d. Select the **Replace if file already exists** check box to replace an existing file of the same name as the pipeline you are exporting.
   e. Optional: Configure your pipeline parameters, if applicable. If your pipeline contains nodes that reference pipeline parameters, you can change the default parameter values. If a parameter is required and has no default value, you must enter a value.
3. Click **OK**.

Verification

- You can view the file containing the pipeline that you exported in your designated object storage bucket.

CHAPTER 6. TROUBLESHOOTING DSPA COMPONENT ERRORS

This table displays common errors found in **DataSciencePipelinesApplication** (DSPA) components, along with the associated status, message, and proposed solution. The **Ready** condition type accumulates errors from various DSPA components, providing a status view of the DSPA deployment.

Type	Status	Error message and solution
ObjectStorageAvailable Ready	False False	Error message: **Could not connect to Object Store: tls: failed to verify certificate: x509: certificate signed by unknown authority** Solution: This issue occurs in clusters that use self-signed certificates with OpenShift AI version 2.9 or later. The AI pipelines manager cannot connect to the object storage because it does not trust the object storage SSL certificate. Therefore, the pipeline server cannot be created. Contact your IT operations administrator to add the relevant Certificate Authority bundle. For more information, see Working with certificates.
ObjectStorageAvailable Ready	False False	Error message: **Could not connect to Object Store Deployment for component "ds-pipeline-pipelines-definition" is missing - prerequisite component might not yet be available. Deployment for component "ds-pipeline-persistenceagent-pipelines-definition" is missing - prerequisite component might not yet be available. Deployment for component "ds-pipeline-scheduledworkflow-pipelines-definition" is missing - prerequisite component might not yet be available.** Solution: The AI pipelines manager might fail to connect to the object storage, and the pipeline server might not be created. Ensure that your object store credentials and connection information are accurate, and verify that the object store is accessible from within the project's associated OpenShift namespace. One common issue is that the object storage SSL certificate is not trusted, particularly if self-signed certificates are used. Verify and update your object storage credentials, then retry the operation.

Type	Status	Error message and solution
ObjectStorageAvailable Ready	False False	Error message: **Wrong credentials for Object Storage: Could not connect to (minio-my-project.apps.my-cluster.com), Error: The request signature we calculated does not match the signature you provided. Check your key and signing method.** Solution: Provide the correct credentials for your object storage and retry the operation.
DatabaseAvailable Ready	False False	Error message: **FailingToDeploy: Dial tcp XXX.XX.XXX.XXX:3306 : i/o timeout** Solution: If the issue persists beyond startup, check for network issues or misconfigurations in the database connection settings.
DatabaseAvailable Ready	False False	Error message: **Unable to connect to external database: tls: failed to verify certificate: x509: certificate signed by unknown authority** Solution: This issue can occur when you use any external database, such as Amazon RDS. The AI pipelines manager cannot connect to the database because it does not trust the database SSL certificate, preventing the creation of the pipeline server. Contact your IT operations administrator to add the relevant certificates. For more information, see Working with certificates.
DatabaseAvailable Ready	False False	Error message: **Error 1129: Host 'A.B.C.D' is blocked because of many connection errors.** Solution: This issue might occur when using an external database, such as Amazon RDS. Initially, the pipeline server is created successfully. However, after some time, the OpenShift AI dashboard displays an "Error displaying pipelines" message, and the DSPA conditions indicate that the host is blocked due to multiple connection errors. For more information on how resolve this issue for an external Amazon RDS database, see Resolving "Host is blocked because of many connection errors" error in Amazon RDS for MySQL. Note: Clicking this link opens an external website.
APIServerReady Ready	False False	Error message: **Route creation failed due to lengthy project name: Route.route.openshift.io is invalid: spec.host exceeds 63 characters.** Solution: Ensure that the project name in OpenShift is less than 40 characters.

Type	Status	Error message and solution
APIServerReady Ready	False False	Error message: **FailingToDeploy: Component replica failed to create. Message: serviceaccount "ds-pipeline-sample" not found.** Solution: If the failure persists for more than 25 seconds during DSPA startup, recreate the missing service account.
PersistenceAgentReady Ready	False False	Error message: **FailingToDeploy: Component's replica failed to create. Message: serviceaccount "ds-pipeline-persistenceagent-sample" not found.** Solution: If the failure persists for more than 25 seconds during DSPA startup, recreate the missing service account.
ScheduledWorkflowReady Ready	False False	Error message: **FailingToDeploy: Component's replica failed to create. Message: serviceaccount "ds-pipeline-scheduledworkflow-sample" not found.** Solution: If the failure persists for more than 25 seconds during DSPA startup, recreate the missing service account.
MLMDProxyReady Ready	False False	Error message: **Deploying: Component [ds-pipeline-scheduledworkflow-sample] is still deploying.** Solution: Wait for DSPA startup to complete. If deployment fails after 25 seconds, check the logs for further information.

6.1. COMMON ERRORS ACROSS DSPA COMPONENTS

The following table lists errors that might occur across multiple DSPA components:

Deployment condition and condition type	Status	Error message and solution
Condition: Component Deployment Not Found Condition type: **ComponentDeploymentNotFound**	False	Error message: **Deployment for component <component> is missing - prerequisite component might not yet be available.** Solution: The deployment for the component does not exist. Typically, this issue occurs due to missing deployments or issues that occurred during creation.
Condition: Deployment Scaled Down Condition type: **MinimumReplicasAvailable**	False	Error message: **Deployment for component <component> is scaled down.** Solution: The component is unavailable as the deployment replica count is set to zero.

Deployment condition and condition type	Status	Error message and solution
Condition: Component Failing to Progress Condition type: **FailingToDeploy**	False	Error message: **Component <component> has failed to progress. Reason: <progressingCond.Reason>. Message: <progressingCond.Message>** Solution: The deployment has stalled due to **ProgressDeadlineExceeded** or **ReplicaSetCreateError** issues, or similar.
Condition: Replica Creation Failure Condition type: **FailingToDeploy**	False	Error message: **Component's replica <component> has failed to create. Reason: <replicaFailureCond.Reason>. Message: <replicaFailureCond.Message>** Solution: Replica creation has failed, typically due to an error in the replica set or with the service accounts.
Condition: Pod-Level Failures Condition type: **FailingToDeploy**	False	Error message: **Concatenated failure messages for each pod.** Solution: Deployment pods are in a failed state. Check the pod logs for further information.
Condition: Pod in CrashLoopBackOff Condition type: **FailingToDeploy**	False	Error message: **Component <component> is in CrashLoopBackOff. Message from pod: <crashLoopBackOffMessage>** Solution: Pod containers are failing repeatedly, often due to incorrect environment variables or missing service accounts.
Condition: Component Deploying (No Errors) Condition: type: **Deploying**	False	Error message: **Component <component> is deploying.** Solution: The component deployment process is ongoing with no errors detected.
Condition: Component Minimally Available Condition type: **MinimumReplicasAvailable**	True	Error message: **Component <component> is minimally available.** Solution: The component is available, but with only the minimum number of replicas running.

CHAPTER 7. ADDITIONAL RESOURCES

- Kubeflow Pipelines 2.0 Documentation